

Applications in Plant Sciences

Are they two seeds in a pod? Comparing seed rain recovery using artificial grass carpets versus sticky traps in grasslands.

--Manuscript Draft--

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Abstract:	Premise. Seed dispersal is a critical process for plant community assembly, however natural rates of seed arrival are rarely quantified when compared to other assembly mechanisms, especially in herbaceous communities. Methods. Here we compare the utility of artificial grass carpet squares ('artificial grass') for capturing seed rain to classic 'sticky trap' methods. We placed paired sticky traps and artificial grass squares in two grassland ecosystems. We added known numbers of seeds of multiple species to each trap, and recovered seeds at one week, one month and two month intervals. Results. We found that both trap types lost seeds through time at similar rates, but each trap type had pros and cons. Overall sticky traps retained more seeds and measured primary dispersal, but recovering seeds was difficult and hindered by debris stuck to the traps. Alternatively, artificial grass traps measured effective dispersal as seeds were lost through time to secondary dispersal and granivory, but recovered seeds could be handled easily and retained for long-term storage and germination. Discussion. We encourage the broad adoption of seed rain studies to increase links between theoretical and empirical community ecology. Both sticky traps and artificial grass traps provide ways to measure seed rain in grasslands, but vary in the types of information they provide.	
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We appreciate the thoughtful comments by the editor and the reviewers. We have revised our manuscript accordingly and have addressed all concerns listed below. Below we describe the major points that we have addressed. We also include a direct response to each comment below.

Main points we have addressed in this revision.

1. Recovery vs capture.

Our goal with this manuscript was to describe a new method for capturing ambient seed rain in grasslands (the artificial turf squares), and compare it to the leading method in the field (sticky traps). The hope is that researchers can use either of these methods to “capture” seeds in the field, which is defined here as putting out the traps and counting/identifying all of the novel seeds that fall onto the carpets. This seed rain capture provides a metric of the seeds falling to the ground. However, if we were to perform capture studies in the field comparing the artificial turf and the sticky traps, we could not ensure we had equal seed fall across all traps to assess capture, and thus an experimental design that focused on capture would not provide the data necessary for researchers to make decisions about which trap types they should use to make sure they were adequately defining capture. Instead, here we perform a “recovery” experiment. Recovery is defined as the number of seeds we are able to successfully recover from the traps after a given amount of time after placing a known number of seeds of each species on each trap. Recovery allows us to validate how likely we are to find and identify *known* seeds on the traps, and how long seeds remain in that location through time. Thus, we need to quantify recovery in order to feel confident in our trap types should they be used for capture studies in the future

We acknowledge that this distinction between recovery and capture was not clear in our previous version of the manuscript. We have made several changes to address reviewer comments:

1. We have defined capture as “These traps are considered to measure “seed capture”, as traps are empty when placed in the field, and seeds that fall into the traps from the surrounding community are captured and measured when traps are removed.” on lines 79-81 and we have gone through the introduction and changed all mentions of recovery to capture when appropriate based on our definitions above.
2. We have included reference to a paper where we quantified seed capture with these carpets “For example, recently Wynne et al. (2024) used artificial grass carpets for capturing seed rain in a successional chronosequence of temperate tallgrass prairies and were able to identify ~27,000-124,000 seeds per square meter depending on the age of the grassland.” on line 107-110 and distinguish this from recovery.

3. We have now defined the importance of a recovery study for determining trap efficacy. On lines 116-121, we say “While our ultimate goal is to use these traps to capture seed rain in the wild, we first need to quantify how well each trap type retains known numbers of seeds under field-conditions. Thus, we conduct “seed recovery” experiments where we put known numbers of seeds on each trap type and see how well we are able to recover these same seeds at distinct time intervals. This recovery measure provides information on how confident we can feel in whether we are measuring *primary* or *effective* seed dispersal.”
 4. In the discussion broadly, we now discuss how sticky traps are more likely to measure primary dispersal because less seeds are lost through time, and artificial turf tarps are more likely to measure effective dispersal as they retain seeds, but allow for ecologically realistic loss to granivory or secondary dispersal.
2. Trap type pros and cons.

We acknowledge that the reviewers both pointed to the importance of simply presenting the recovery data on trap types, and letting future trap users decide which method makes the most sense for their system. We painfully agree with this comment. While we feel the sticky traps are unfortunately disgusting and hard to work with, we agree it is very important to remain unbiased in our manuscript. We have now presented this manuscript in a non-judgemental way that allows readers to decide which method makes sense to them. However we leave reviewers with this quote from a recent PhD thesis we found that used the sticky trap method because it sums up our feelings about sticky traps.

“Sticky taps are a pain to work with; extracting and identifying the seeds from the glue, ensuring that the glue remains sticky while in the field, etc.” Asefa, G. (2011) PhD Thesis: The dynamics of coexistence in annual plants. Imperial College, London.

Editor -----

Comment: Editor-in-Chief: Thank you for this useful submission to the special issue (and thank you for your work as an editor!). I think this will be a very useful study for our colleagues as they seek to better understand seed dispersal going forward. I enjoyed read your manuscript, and found it generally very clear and approachable. That being said, I also think the reviewers have offered extremely useful comments that will help you differentiate between seed capture and seed retention and better communicate the utility of the different seed traps. I look forward to seeing your revised manuscript come across my desk.

Response: Thank you and we have addressed the reviewer concerns here!

While you are revising the manuscript according to the reviewer comments, please also take a moment to complete the following editorial changes:

Comment: - L105-114: Combine these sentences into a single paragraph.

Response: Done!

Comment: - As per our author guidelines, please give the taxonomic authority for each species on first mention (or you could add them to Table 1).

Response: We have now added taxonomic authority (as based on Flora of North America) to Table 1.

Comment: - Please note that supplemental information in APPS articles is referred to as a "supplemental appendix" - this will be a very easy change for your one supplemental figure.

Response: corrected!

Comment: - Double-check your github link to make sure the data is accessible.

Response: fixed!

Reviewer 1 -----

Comment: The authors present the results of a well-replicated experiment in two types of grassland to evaluate two types of methods for estimating seed rain. Overall, the paper is well written, the statistics are solid, and the figures are clear (the use of the photos in Figure 1 was great!). I have just a few suggestions for clarifying the scope - in particular, clarifying the idea of seed recovery and how that might differ from actual seed dispersal in grasslands.

Response: Thank you!! We appreciate it!

Comment: 1. I found myself most confused by the idea of seed recovery - in some places it's very clear the authors are comparing these two methods, but in others, the proportion of seeds in the trap led me to first think about the proportion of seeds a trap would catch from natural seed rain - and I know it isn't that. To make the paper more useful - it would be worth adding something short in the introduction and discussion - how do we interpret these results - for example, how well do traps catch seeds compared to the soil? (I get there are no data, but worth a speculation). Otherwise, it feels so specific as to be somewhat less useful.

Response: Please see Main Point 1.

Comment: 2. Related to this first point, the word 'recovery' could be better defined early on. E.g. at Line 93 - 'recovery' is how many seeds stay in the trap that were put in the trap - but how many would get to the trap in the first place? Can recovery be defined more clearly in this paragraph? (Comes up again at L129 - seeds were placed in the trap, but maybe more than would actually get there on their own? Use previous studies to compare numbers?) The first paragraph of the discussion uses the phrase 'seed recovery efficacy' - this is wordier, but also more precise to me.

Response: Thank you for this suggestion, which we have incorporated. Please see Main Point 1. Additionally, we now mention in the methods that the added species were not found in any of the vegetative communities of the site, so ambient arrival was impossible. This allows us to ensure that all seeds were from recovery of those added rather than capture of seeds that might have dispersed from the nearby community.

Comment: 3. In the methods & results - I would have liked to read about absolute seed recovery first, compared to relative seed recovery - if both are equally bad for a particular species, for example, then relative seed recovery will be the same, but actually, neither method is that suitable.

Response: Thank you for this point. There was one species that had low recovery for both trap types (*Helianthus annuus*) but all others either had similar recovery, or it varied by species so this information is valuable for readers. We have now added a new figure to the manuscript (new Figure 2) and discuss the results on lines 210-213 "Absolute and relative seed recovery varied by species and time (Figs. 2,3). There was high variability among absolute recovery by species, with some species having higher recovery broadly (e.g. *S. scoparium*, *B. dactyloides*), others had low recovery across all trap types (e.g. *H. annuus*), but most falling somewhere in the middle."

Comment: 4. In the results, I would avoid the word 'caught' - this implies that the traps caught seeds that were naturally dispersed. 'trapped' or 'retained' instead?

Response: this has been adjusted (see Main Point 1)

Comment: 5. Do you have any advice for users as to how long the traps should stay out, given that in nature not all species disperse their seeds at the same time?

Response: In the same way we don't want to promote one trap over another, we also don't want to dictate how often traps should be replaced given the many issues experimentalists face with deciding when and how often they can go to the field. On line 340-342 we now say

“Deciding how often to replace seed rain traps can often be dictated by person-power and ability to get to field sites. Therefore, we suggest replacing traps at a rate that is feasible and interpreting results with caution given seeds are lost through time on all trap types.”

Comment: 6. Table 1 might be more useful if species were ordered from lightest to heaviest within each site.

Response: We have re-organized the table to be sorted by seed size.

Comment: 7. Fig 3 - make the y-axis range from 0 to 1 to emphasize the efficacy further, perhaps also for Fig 4, but definitely for Fig 3. I also wondered if points were jittered - if so, be clear in the figure legend.

Response: We have altered the figure as suggested.

Comment: 8. Note: the GitHub link provided with the manuscript did not work for me, but I was able to get there through the parent directory for Sullivan-Lab.

Response: Thanks for catching this! We've updated the link.

Reviewer 2 -----

Comment: The manuscript APPS-D-25-00043 titled "Are they two seeds in a pod? Comparing seed rain recovery using artificial grass carpets versus sticky traps in grasslands" presents a comparison between two methods to assess seed rain in different grassland types. The authors show that sticky traps retain more seeds, especially smaller ones, but need to be replaced more frequently and the retrieved seeds cannot be used for germination, limiting their use for unidentified seeds. The topic is relevant and has clear applications, but I had a couple of concerns with the current presentation. I also had a few minor comments that I hope will be helpful to the authors when revising their manuscript.

Response: We thank the reviewer for their suggestions, which we have implemented as described below.

Comment: 1) Seed rain capture vs retention: I am having a bit of trouble in seeing whether your methods allow assessing seed capture or seed retention in the traps (or both?). You use these terms sometimes interchangeably in the text (e.g. seed rain capture methods are mentioned in the end of the introduction in L100, but seed retention is specifically mentioned in the

description of the aims of the study in L91 and in the methods in L116), and then some of the results mention specifically seeds caught (L194, 196). I guess this depends on whether you aimed at simulating seed rain on the traps, but I understand this was not the case, as you mention that you "placed" seeds on the traps (L129). If this is the case, you cannot really assess seed capture, so you are missing an important part in the assessment of seed rain.

Response: We have clarified this point; please see Main Point 1 for more details.

Comment: Overall message: given that your results show advantages and disadvantages for each type of seed trap and not overwhelming support for a particular method, I am not convinced you can encourage researchers to use one method over the other one, as you do now in the abstract and the beginning of the discussion. I think the main point here is to make researchers aware of the limitations and advantages of each method, and let them decide, depending on their research questions. You provide a more balanced view in some parts of the discussion (e.g. L292-294), and I think that should be the main message to convey. Possibly an effective way forward could be to suggest using a combination of methods, where artificial grass traps are used as the main method for seed rain assessment, but sticky traps are placed for a short period (during peak seed production?) to at least have some estimate of how much of the seed rain, especially of the smaller seeds, you are missing?

Response: We have now changed the discussion to be non-judgemental to trap type to allow readers to decide for themselves which method makes sense for their question (see Main Point 2)

Comment: L5: here you emphasize the importance of being able to retrieve small seeds to be able to more accurately estimate rates of seed arrival in herbaceous communities, then you show that sticky traps were actually better in recovering small seeds (L13), but still you encourage the use of artificial grass (L17), which seems a little contradictory. In your discussion you provide a more balanced view on the advantages and disadvantages of each method, which I think should also be reflected in the abstract.

Response: We have altered this line so that we do not advocate for one method over another (see main point 2 for more info)

Comment: L7-8: is the mention of Arctic systems necessary here? For me it opened more questions - was the use of artificial grass successful in Arctic systems? Was the comparison between sticky traps and artificial grass conducted there too? Has artificial grass not been used in systems other than Arctic? You answer many of these questions in the main text, so I see why you had this sentence here, but I found it a little bit confusing in the abstract.

Response: Thank you for this point. We have now removed this mention from the abstract. We just didn't want readers to think we were the first to think of this idea. Given the confusion, we removed it from the abstract and save this detail for the Introduction.

Comment: L18: the ending of the sentence seems too broad- you explain this better in the main text, but it feels a bit out of context here

Response: We updated this to now read: "We encourage the broad adoption of seed rain studies to increase links between theoretical and empirical community ecology. Both sticky traps and artificial grass traps provide ways to measure seed rain in grasslands, but vary in the kinds of information they provide." This was also requested by Reviewer 1

Comment: L22: should this be "plant communities"? vegetative seemed to contradict seed dispersal ;)

Response: Yes! Thank you. This is now changed.

Comment: L95: use "artificial grass traps" here for consistency?

Response: perfect thanks for catching this. We have changed this here and throughout.

Comment: L126: how far apart were the sticky traps and the artificial grass traps? Were they spatially paired (i.e. closer to each other than to the other replicates in a plot)?

Response: Yes, we have now added this for clarity.

Comment: L133: can you report the size of Yucca seeds here?

Response: we have now included this.

Comment: L137: I understand from this statement that you recorded only "recaptured" seeds - was there no natural seed rain?

Response: We did have natural seed rain onto the traps. We have information on this but did not include it in the paper for simplicity, so that we can focus on recovery.

Comment: L165: can you include site as a random factor if it only has two levels, or would this be better included as a fixed factor in the mixed effects model? Also because the two sites

represent two different habitats (tallgrass vs shortgrass) it could be useful to have that as part of the main question as a fixed effect.

Response: This is a stats philosophy question for sure, and you can run these models either way. Because we only have two sites, we don't feel comfortable making statements about these ecosystems specifically and would prefer to talk about the patterns broadly - thus leaving them as random effects. However, we tried running these models with site as a fixed effect and there are different levels of recovery between the two sites. This is accounted for with site-level random intercepts now, but we would love to know how these rates of recovery vary across a larger number of sites and site factors!

Comment: L199: were there differences in seed loss between trap types? Maybe you could indicate here the observations of seed fragments that could be related to seed predation?

Response: there were not differences in seed loss per trap type per time. We did add the reference to the seed fragments and possible predation.

Comment: L217-220: see my comments above - given some of the limitations of artificial grass traps you mention, I think you should (definitively!) provide the reasons, but let the readers decide by themselves ;)

Response: We agree! We have now made this change (see Main Point #2 for details)

Comment: L224: does this then mean that artificial grass traps had 48% seed loss?

Response: yes, on average

Comment: L236: but maybe some of those non-recovered seeds are just washed away from the traps but stay close by and still contribute to the aboveground community? (not sure if this question makes a lot of sense?)

Response: Yes, they totally could have! Just not in the specific location where the trap is. We've updated the statement to better reflect our original intention of discussing effective dispersal, after accounting for granivory or secondary dispersal.

Comment: L237-239: not sure I follow the logic here. Why is this only for artificial grass traps?

Response: It's true that some seeds were lost for sticky traps, just a lot less. So that's more of a measure of primary dispersal on sticky traps because if you assume anything that could get

lost to the environment, does in artificial grass traps (because they are open to the environment) and so that's the lower limit of what you'd expect in the area, then the sticky traps are a LOT higher. It's true they are not a perfect measure of primary dispersal, but they are a lot closer than the astroturf carpets.

Comment: L269: did you find evidence of artificial grass traps continuing to collect and store seeds?

Response: Good question. We did find that the astroturf carpets collected seeds that we did not initially place on the carpets

Comment: L280: "granivores"

Response: corrected!

Comment: L285: "...are more..."

Response: corrected!

Comment: L292-294: nice! :)

Response: thanks!

Comment: L303: totally agree!

Response: great!

Comment: L304: what do you mean by a "wider range of conditions"? You did not really measure this

Response: We meant that the carpets can handle more weather conditions (e.g. still capture seeds after freezing). We have adjusted the text to be more specific and now say: "Artificial grass traps are effective at seed recovery under a wider range of weather conditions (e.g. can capture seeds post-frost)..."

Comment: L308: it might be good to include a figure (maybe in supplementary?) showing the proportion of (total?) seeds recovered in each trap type over time, to support this statement.

Response: Yes! Good point. We did have that figure in the supplement but we did not cite it here, so it was initially easy to miss. We now include a reference to supplementary figure S1.

Running head: Sullivan et al. Grassland seed rain methods comparison

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Are they two seeds in a pod? Comparing seed rain recovery using artificial grass carpets ~~earpets~~ versus sticky traps in grasslands.

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~~September 17, 2025~~

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Keywords: seed arrival, plant ecology, community ecology, methods, artificial grass carpet,
sticky traps, seed rain, grasslands

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ABSTRACT

Premise. Seed dispersal is a critical process for plant community assembly, however natural rates of seed arrival are rarely quantified when compared to other assembly mechanisms, especially in herbaceous communities.

Methods. Here we compare the utility of artificial grass carpet squares ('artificial grass') for capturing seed rain to classic 'sticky trap' methods. We placed paired sticky traps and artificial grass squares in two grassland ecosystems. We added known numbers of seeds of multiple species to each trap, and recovered seeds at one week, one month and two month intervals.

Results. We found that both trap types lost seeds through time at similar rates, but each trap type had pros and cons. Overall sticky traps retained more seeds and measured primary dispersal, but recovering seeds was difficult and hindered by debris stuck to the traps. Alternatively, artificial grass traps measured effective dispersal as seeds were lost through time to secondary dispersal and granivory, but recovered seeds could be handled easily and retained for long-term storage and germination.

Discussion. We encourage the broad adoption of seed rain studies to increase links between theoretical and empirical community ecology. Both sticky traps and artificial grass traps provide ways to measure seed rain in grasslands, but vary in the types of information they provide.

~~**Premise.** Seed dispersal is a critical process for plant community assembly, however natural rates of seed arrival are rarely quantified when compared to other assembly mechanisms, especially in herbaceous communities. This discrepancy arises from the difficulty in recovering small seeds.~~

~~**Methods.** Here we compare the utility of artificial grass carpet squares ('artificial grass') for capturing seed rain to classic 'sticky trap' methods. While artificial grass has been used in Arctic systems, we examine its efficacy in temperate grasslands. We placed paired sticky traps and artificial grass squares in two grassland ecosystems. We added known numbers of seeds of~~

multiple species to each trap, and recovered seeds at one week, one month and two month intervals.

Results. We found that while both trap types lost seeds through time at similar rates, sticky traps recovered more small seeds while artificial grass recovered more large seeds. Overall sticky traps retained more seeds, but recovering seeds was difficult, and recovery was hindered by debris stuck to the traps. Alternatively, as to secondary dispersal and granivory seeds recovered from artificial grass could be handled easily and retained for long term storage.

Discussion. We encourage artificial grass traps for broad adoption of measuring seed rain in grasslands to increase links between theoretical and empirical community ecology.

INTRODUCTION

Dispersal is a critical process that structures populations and communities, linking local patch-scale dynamics to regional landscape-level processes (Amarasekare, 2003; Snyder and Chesson, 2003; Leibold and Chase, 2018). Within ~~vegetative-plant~~ communities, seed dispersal is one of the primary mechanisms of arrival to new habitats, allowing species to track changing environmental conditions and promoting landscape connectivity (Cubiña and Aide, 2001; Simberloff, 2009; Beckman and Sullivan, 2023). Dispersal also plays a critical role in assembly and long-term diversity patterns, such as maintaining source-sink dynamics (Mouquet and Loreau, 2003; Craig et al., 2025), allowing long-term species co-existence (Snyder and Chesson, 2003; Berkley et al., 2010), and maintaining genetic diversity in populations (Jordano, 2017). A species' dispersal ability also has direct management implications, often increasing native diversity after restoration, or alternatively, facilitating the invasion of non-native species (Von Der Lippe and Kowarik, 2007; Brudvig et al., 2009; Sperry et al., 2019). Despite its importance as a critical process for populations and communities (Schupp and Fuentes, 1995; Levine and Murrell, 2003; Vellend, 2010), dispersal is notoriously difficult to quantify empirically (Bullock et al., 2006; Beckman and Sullivan, 2023). These methodological challenges have created a gap between our theoretical understanding of the importance of dispersal versus our empirical understanding and use in applied settings (Bolker et al., 2003), making dispersal arguably the least empirically understood mechanism of community assembly (HilleRisLambers et al., 2012; Vellend, 2016).

One way to measure seed dispersal is by estimating seed rain, or the seeds that fall into a given area (Arruda et al., 2018). While this method ~~captures-quantifies~~ only a single point in the dispersal trajectory of a seed (the arrival stage; Travis et al., 2012), it is arguably the most important phase to understand for plant community ecology, as it provides information on the number and type of seeds that arrive to a specific location. If seeds are recovered before post-

dispersal seed predation or secondary dispersal occurs, then this is considered a measure of *primary dispersal*. However, if seeds that fall to a given area are then lost, ~~to these processes~~ this measure of seed arrival is considered to be *effective dispersal* as it constitutes the seeds that have a chance to germinate and establish in a given area (Nathan and Muller-Landau, 2000). Quantifying seed rain as a measure of seed arrival across many different natural environments is critical for building strong links between plant community ecology experiments and theory (Myers and Harms, 2009).

Seed rain is typically measured through the use of seed traps placed on or near the ground in an ecosystem of interest (Kollmann and Goetze, 1998; Cottrell, 2004). Seed traps are commonly used in forest systems to measure seed rain (Cottrell, 2004; Rogers et al., 2017; Piotto et al., 2019; Huanca Nuñez et al., 2021), and when there are known parents, dispersal trajectories can be calculated using inverse modeling (Nathan and Muller-Landau, 2000; Wälder et al., 2009). These traps are considered to measure “seed capture”, as traps are empty when placed in the field, and seeds that fall into the traps from the surrounding community are captured and measured when traps are removed. Seed rain measurements, however, are much less common in ecosystems with ~~predominantly~~predominantly herbaceous communities (e.g. grasslands, tundra), with only a handful of studies estimating seed rain in grasslands specifically (Rabinowitz and Rapp, 1980; West and Durham, 1991; Schott and Hamburg, 1997; Kettenring and Galatowitsch, 2011; Willand et al., 2015; Wynne et al., 2024). This dearth of knowledge on herbaceous seed rain likely occurs because ~~recovering capturing~~ small herbaceous seeds is challenging (Myers and Harms, 2009) and seed identification resources for even common species’ can be scarce (Wynne et al., 2024). The main method for capturing seed rain in grasslands is via sticky traps, where a flat surface (often a ~~petri~~Petri dish) is sprayed with a sticky substance, and seeds that falls onto the surface are captured and later recorded (Rabinowitz and Rapp, 1980). Sticky traps present multiple practical problems, including that they ~~have need~~ to be replaced frequently because the sticky surface can become completely covered in leaves and debris, thereby blocking

newly fallen seeds from capture. Furthermore, sticky traps are no longer sticky once temperatures fall below freezing, despite this being a critical time for seed dispersal in temperate grasslands (Rabinowitz and Rapp, 1980). Lastly, using ~~recovered~~captured seeds from sticky traps for additional work, such as germination trials or trait measurement, is rarely possible as seeds become coated with the sticky residue.

An alternative method for measuring seed rain is to use artificial grass carpets. With this method, artificial grass patches are fixed to the ground, capturing seeds that fall and become lodged in the ~~seeds are dispersed into traps, and they are caught in~~ artificial grass strands. This method ~~is~~has previously been used to sample seed rain in the Arctic (Ulf Molau, 1996; Molau and Larsson, 2000), and provides a potential solution to the many issues that arise with collecting seed rain via sticky traps in grasslands more generally. Artificial grass can remain in the field for long periods of time and still capture seeds, withstand large temperature and hydrologic fluctuations, and collected seeds can be germinated after capture should visual identification not be possible due to lack of seed identification resources (Ulf Molau, 1996; Wolters et al., 2004). These features make artificial grass ~~carpets-traps~~ a viable option for seed rain studies in grasslands. For example, recently Wynne et al. (2024) used artificial grass ~~carpets~~carpets for capturing seed rain in a successional chronosequence of temperate tallgrass prairies and were able to ~~recover~~identify ~27,000-124,000 seeds per square meter depending on the age of the grassland. Although the artificial grass carpet method appears promising for widespread use for collecting seed rain in grasslands, we currently lack a comparison between this and the sticky trap method to compare their ability to ~~recover~~capture seeds and to consider potential biases across methods.

Here, we investigate the effectiveness of traditional sticky traps (Tangle-Trap® resin on Plexiglass® Petri plates) as compared to artificial grass traps for measuring seed rain in grasslands. While our ultimate goal is to use these traps to capture seed rain in the wild, we first need to quantify how well each trap type retains known numbers of seeds under field-

conditions. Thus, we conduct “seed recovery” experiments where we put known numbers of seeds on each trap type and see how well we are able to recover these same seeds at distinct time intervals. This recovery measure provides information on how confident we can feel in whether we are measuring *primary* or *effective* seed dispersal. ~~We examined the seed retention efficacy of both methods through time, comparing the proportion of seeds recovered across two temperate grassland ecosystems, a diversity of species, and a range of seed sizes. If we recover less seeds, we consider this effective dispersal. We examined the seed retention efficacy of both methods through time, comparing the proportion of seeds recovered across two temperate grassland ecosystems, a diversity of species, and a range of seed sizes.~~ While traditional sticky traps had a higher recovery rate of smaller seeds, both methods recovered a decently high proportion of seeds overall, and seeds from artificial ~~turf-grass~~ traps were in better condition for counting, identification, and post-processing measurements. As such, we suggest that sticky trap capture data are more likely to measure primary dispersal, whereas artificial ~~turf-grass~~ trap capture data are more likely to measure effective dispersal. s be used for measuring seed rain in grassland communities, especially when it may be beneficial to germinate collected seeds or in remote locations where it is difficult to replace traps frequently (Rabinowitz and Rapp, 1980; Cottrell, 2004). By having a robust comparison s of different seed rain capture methods, we can better advance our empirical understanding of dispersal, ~~thus~~ narrowing the current gap between theory and data, and advancing our understanding of the role of dispersal in grassland restoration and conservation.

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METHODS

Study sites ~~and seed mixes~~

We replicated experimental tests of seed capture methods for detecting seed rain in two distinct grassland communities: a shortgrass prairie in northern Colorado, and a tallgrass prairie in central Missouri (Fig. 1A, B).

The Colorado (CO) study occurred on private land bordering the Cathy Fromme Prairie Open Space, a 267 ha remnant of native shortgrass prairie. The site is relatively flat, surrounded by mature cottonwood trees, and receives approximately 38 cm of precipitation annually (Balok et al., 1995).

The Missouri (MO) study occurred on a restored tallgrass prairie owned by the University of Missouri. The prairie was located on campus, and is surrounded by buildings and ~~foot traffic~~foot traffic, and receives approximately 108 cm of annual precipitation (Gu et al., 2015).

Experimental design ~~and tests of trap performance~~

We tested the effectiveness of sticky traps versus artificial grass carpets for seed retention through time. Following standard protocol (Rabinowitz and Rapp, 1980; Kollmann and Goetze, 1998; Chabrierie and Alard, 2005), sticky traps consisted of 10 cm x 10 cm Plexiglass plates coated with Tangle-Trap® sticky coating. To compare this standard method to ~~seed carpets~~artificial grass traps, we used 10 cm x 10 cm synthetic lawn turf squares (TrafficMASTER® TruGrass Emerald Artificial Turf) with blade heights between 1.5 to 2.5 cm tall.

At the two sites, we replicated 10 blocks of paired sticky ~~traps~~ and ~~carpets~~artificial grass traps that we collected 1 week, 1 month, and 2 months after deployment to examine seed recovery through time. We initially deployed all traps on July 10, 2019 in Colorado and July 17,

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2019 in Missouri. Each block consisted of three sticky traps and three ~~carpets~~ artificial grass
traps for a total of 10 replicates of ~~the two each trap type methods~~ for each ~~each~~ time point and
each site. Within each location block, paired traps were placed side by side, closer to each other
than the other replicates in other blocks. When deploying traps, we removed any litter or
vegetation directly under the traps so that they were placed directly on the soil surrounded by
natural vegetation (Fig. 1E). We used two ground staples to secure opposite sides of each trap.
On each trap, we then placed 20 seeds of 8 locally occurring species that were not found in the
neighboring vegetative community at our study sites. ~~where~~ Seed mixes were chosen such that
they encompassed a range of functional groups and seed mass (see Table 1 for species and
associated seed mass, Fig. 1C, D). At the Colorado site, we used only 10 seeds for the species at
the highest end our seed mass range, *Yucca glauca*, due to their large size (~7x10mm, Fig. 1D),
resulting in a total of 150 (CO) and 160 (MO) seeds per trap.

We collected paired sticky and artificial grass trap ~~traps and carpets~~ 1 week, 1 month,
and 2 months after deployment. All traps were taken back to the lab for processing. We recorded
the total number of seeds of each focal species that were recaptured on each trap, and noted if
seed fragments were evident to suggest seed herbivory or animal activity.

At the Missouri site, we also wanted to test the efficacy of the sticky traps after they had
been in the field for some time and accumulated leaves and other debris. Therefore, at the initial
trap deployment date, we placed an additional sticky trap at each block that was coated in
Tangle-Trap®, but did not have the seed mix added initially. After one month of sitting in the
grassland and accumulating leaves and other debris naturally, we added the same seed mix as
above to these traps, and then collected these traps along with the two month traps. We termed
this treatment, the ‘post-sticky’ treatment. We recovered and identified seeds from these traps in
the same manner as mentioned above. This allowed us to more closely examine how well sticky
traps recovered seeds after longer periods of time in the field.

Statistical analysis

We explored both absolute and relative seed recovery. To visualize how relative seed recovery varied across species and trap type, we calculated log response ratios of seed recovery of sticky traps relative to artificial grass ~~carpet~~-traps for each species at each site (i.e. the log of % recovery on sticky traps divided by % recovery on artificial grass traps). For this formulation, values greater than zero indicate seeds are more likely to be recovered on sticky traps, values of less than zero are more likely to be recovered on ~~carpets~~grass traps, and values of zero indicate seeds are equally likely to be recovered on both traps. This allows for ease of visualization of which trap type tends to recover relatively more seeds of a given species at a given time.

To determine how absolute seed recovery depended on the trap type, seed mass and the amount of time the traps remained in the field, we used generalized linear mixed effects models using the ‘glmmTMB’ package (Brooks et al., 2017) in R v4.3.3. (R Core Team, 2024). The percent of recovered seeds was our response variable, and accordingly we used a binomial model with weights specified as the number of seeds placed on the traps to account for differing numbers of seeds per species used in the experiment. Our predictor variables included the interaction of trap type, seed mass and time in the field. We also included an additive random intercept for block nested within site (MO or CO). We used a backward selection approach to determine which interactions we kept in the model, and removed non-significant interactions when they arose. We calculated contrasts between trap type predictions for given seed masses using ‘emmeans’ (Lenth, 2023).

To determine how well the sticky traps recovered seeds once the traps had been in the field for some time and acquired debris, we explored how recovery differed between the poststicky treatment and the other trap types that contained seeds for one month. At the Missouri site only, we subset our data to all traps that had seeds on them for one month (artificial grass, sticky and post-sticky). We then ran a similar generalized linear mixed effects model as above,

231 with our response variable as the proportion of seeds recovered (modeled as a binomial
232 distribution), our predictor variable as trap type, and two additive random intercepts, one for
233 block and one for species. We then calculated contrasts with ‘emmeans’ between trap type.

234 We wrangled and visualized data with the ‘tidyverse’ package (Wickham et al., 2019),
235 and ran all code in R Studio v2024.12.0+467 (Posit Team, 2024).

236 RESULTS

237 ~~Absolute and r~~Relative seed recovery varied by species ~~and time and time~~ (Fig. 2Figs.
238 2,3). ~~There was high variability among absolute recovery by species, with some species having~~
239 ~~higher recovery broadly (e.g. *S. scoparium*, *B. dactyloides*), others had low recovery across all~~
240 ~~trap types (e.g. *H. annuus*), but most falling somewhere in the middle.~~ We also observed that
241 some species were more likely to be recovered on sticky traps (positive log response ratio, e.g. *C.*
242 *tinctoria*, *D. purpurea*), others were more likely to be recovered on artificial grass traps (negative
243 log response ratio, e.g. *H. annuus*, *S. heterolepis*), and many were equally likely to be recovered
244 on either trap type (e.g. *S. scoparium*, *B. dactyloides*). These trends often varied with time.

245 We next explored how trap type, seed mass, and the time the traps spent in the field
246 altered the absolute proportion of seeds recovered. We found a significant interaction between
247 seed mass and trap type, and a significant additive effect of time (Table 2). On average, sticky
248 traps recovered more seeds than artificial grass traps, with sticky traps recovering $\sim 74\% \pm 13\%$
249 ~~SD~~ of seeds, and artificial grass traps recovering $\sim 52\% \pm 18\%$ ~~SD~~ of seeds. However, this
250 relationship depended on seed mass (Fig. 3Fig. 4). As seed mass increased, the proportion of
251 seeds recovered increased on artificial grass ~~carpet~~traps, but decreased on sticky traps. For small
252 seed masses, artificial grass traps caught significantly fewer seeds (log odds estimate at seed
253 mass of 0, estimate = -1.56 ± 0.04 SE, $p < 0.001$). However, at large seed sizes, artificial grass
254 traps caught significantly more seeds (log odds estimate at seed mass of 0.02, estimate = $0.75 \pm$

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0.13 SE, $p < 0.001$). Additionally, the longer the traps were in the field, the more seeds were lost (Supplemental Information Fig. S1). Across both trap types, ~ 30% seeds were lost for each month the traps spent in the field. This loss was due in-part to seed predation as we noted seed fragments left on both trap types post-consumption.

Finally, to determine how well the sticky traps continue to capture seeds after some time in the field, we conducted the ‘post-sticky’ trial at the Missouri site. We found that sticky traps caught a significantly larger proportion of seeds than the post-sticky treatment, but there was no difference found between the seeds captured with the ~~artificial~~ artificial grass and the post-sticky treatments (Fig. 4 Fig. 5, Table 3). The post-sticky traps captured ~ 30% less seeds than the sticky traps.

DISCUSSION

A broader understanding of the natural rates of seed rain in grasslands is necessary to understand the role of seed arrival for the assembly and reassembly of plant communities under global change (Myers and Harms, 2009; Arruda et al., 2018). Standardized methods for estimating seed arrival and dispersal dynamics that are flexible and can adequately recover seeds under a myriad of conditions is critical, given that grasslands span ~ 40% of the Earth’s surface, and occur on all continents across a wide range of precipitation and temperature regimes (Buisson et al., 2022). Here we provide a test of the seed recovery efficacy of two methods for measuring seed rain in grassland communities: artificial grass traps and sticky traps. These methods are both easy to deploy because of their size and minimal impact on the local community (as compared to other funnel or pitfall-based methods; Kollmann and Goetze, 1998). While both methods are successful for recovering seeds, our experimental comparison highlights that sticky traps are more likely to represent primary dispersal because they lose fewer seeds through time, whereas the artificial grass traps represent effective dispersal because

fewer seeds remained at each point and thus had fewer opportunities for future germination. ~~has led us to encourage the use of artificial grass traps for capturing seed rain in grasslands, especially for seeds that are larger in mass, for the reasons we describe below.~~ Obtaining estimates of local primary and effective dispersal across a range of grassland communities will help bridge the gap between dispersal theory and empiricism (Amarasekare, 2003; Bolker et al., 2003; Myers and Harms, 2009).

We find that both the artificial grass and sticky trap methods have high rates of seed recovery (72% for sticky traps on average, compared to 52% for artificial grass traps), with decreasing recovery through time (~30% seeds lost per month overall). Seed loss from both trap types was ~~due to both abiotic and biotic reasons, such as strong rainfall events washing seeds off carpets the artificial grass or herbivory of seeds.~~ In our study, we found evidence for both scenarios; for example a large, single day rainfall event decreased seed recovery at the Colorado site between time point 1 (1 week after deployment) and time point 2 (1 month after deployment). Evidence of both insect and small mammal herbivory occurred across both sites, with hundreds of ants on some sticky traps (Fig. 1G), cottontail rabbit (*Sylvilagus sp.*) feces on artificial grass seed carpets (Fig. 1F), and husks of seeds post-consumption ~~were~~ found on both trap types, especially after longer periods of time. These findings indicate that no trap type is a perfect representation of primary dispersal, but the sticky traps come the closest. Alternatively, the artificial grass traps but instead represents a measure of *effective dispersal* (Nathan and Muller-Landau, 2000), where recovered seeds are those that have escaped herbivory-granivory or secondary dispersal and could potentially germinate and join-contribute to the aboveground community. Therefore, the artificial grass trap method arguably represents more “ecologically relevant dispersal” for the local community, because the traps allow for seed loss that would occur naturally on the soil surface - leading to recovered seeds being those that have the potential to germinate. We suggest future studies examine herbivory rates on ~~seed carpets~~artificial grass traps, deploying paired ~~carpets-traps~~ in herbivore removal and control

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plots ~~to estimate the levels of biotic seed removal. Artificial grass trap~~ ~~Seed carpet~~ methods could be extended from our current focus on seed rain to examine herbivory rates or even conduct preference experiments in grasslands (Edwards and Crawley, 1999; Bakker and Olff, 2003).

While sticky traps are more commonly used in field studies (e.g. Rabinowitz and Rapp, 1980) and retain more seeds on average, ~~they also have many drawbacks for accurately measuring seed rain~~ our results demonstrate these traps have some drawbacks. Sticky traps capture everything that comes in contact with the surface of the ~~Petri dish~~plate (Fig. 1G), and thus sorting seeds from plant and soil debris as well as insect bodies can be difficult. Recovery is especially difficult for small seeds that have the potential to stick to debris in the trap and be easily missed. In our work we were focused on finding a known set of species. However, once the identity of seeds are unknown (as in classic seed rain ~~capture~~ studies, compared to our comparative methods-focused ~~recovery~~ approach), the ability to lose seeds due to accidentally sticking to other material will decrease seed ~~recovery~~capture. We also noticed that some species with thinner seed coats changed color after being in contact with Tangle-Trap® for long periods of time, which can hinder identification if seeds are not removed and identified quickly. Given how time-consuming seed ~~removal~~capture and identification can be (Wynne et al., 2024), it is critical to have a method that allows samples to be stored for long periods and remain unaffected. In addition, it is nearly impossible to ~~recovery~~remove seeds from sticky traps and then germinate them to determine viability or identity, whereas this is quite simple with artificial grass ~~carpet~~traps. The potential to germinate ‘unknowns’ is, in our view, a key advantage of artificial grass traps, given unknowns are extremely likely in seed rain studies (Wynne et al., 2024). Finally, the timing of sticky traps is extremely important for their effectiveness. Sticky traps no longer capture seeds effectively once they have been frozen (Ulf Molau, 1996), which makes post-frost seed rain collection impossible despite many temperate seeds dispersing late in the growing season or under snowy conditions (Rabinowitz and Rapp, 1980; Molau and

331 Larsson, 2000). In addition, our results from the Missouri site clearly show that sticky traps
332 work best when they are in the field for short periods of time, or else debris collects on the trap
333 and hinders seed capture and recovery (Fig. 4 Fig. 5). This is in comparison to artificial grass
334 traps that continue to effectively collect and store seeds regardless of time or temperature. Our
335 work strongly supports the use of artificial grass traps as a primary method to measure seed rain
336 in grasslands because these traps improve the ease of identifying unknown seeds, and can
337 successfully recover seeds under many different site conditions and time scales.

338 Our results demonstrated that the relationship between seed mass and seed recovery
339 changes with trap type (Fig. 3 Fig. 4). As seed size increases, we see a small decrease in the
340 proportion of seeds recovered on sticky traps, but a large increase in the proportion of seeds
341 recovered on artificial grass traps. This result likely arises from differences in the body size of
342 the granivores consuming seeds from the traps. Broadly, there is a positive relationship between
343 granivore body size and the size of seeds they consume (Muñoz and Bonal, 2008), with smaller-
344 bodied insect granivores tending to favor smaller seeds than larger-bodied small mammal
345 granivores granivores (Rey et al., 2002; Radtke, 2011; Vaz Ferreira et al., 2011; Maron et al.,
346 2012). The loss of smaller seeds from the artificial grass traps likely reflects how well the
347 Tangle-Trap® captures insects like ants that attempt to remove seeds from these traps, but instead
348 become stuck to the surface of the sticky trap (Fig. 1G). Therefore, more smaller seeds remain
349 to be recaptured on the sticky traps because insect granivores are not removing them.
350 Conversely, larger seeds are more likely to be recovered on the artificial grass traps. Here,
351 larger seeds are likely consumed by small mammals, and are likely easier to consume from
352 sticky traps because they are fixed on the soil surface and not hidden among the strands of grass
353 as they are in the artificial grass traps. This decrease in handling time could increase
354 consumption rates of larger seeds from sticky traps. While the exact reason for why we find this
355 interaction between trap type and seed mass should be further explored, this result underscores
356 the importance of considering the difference in the results provided by the different trap

357 methods. If the goal is to understand primary seed dispersal and to quantify the number and type
358 of seeds that fall regardless of their fate, we encourage the use of sticky traps that are replaced
359 very frequently (and of course considering all possible concerns listed above). However, if the
360 goal is to understand ¹-effective dispersal², which focuses more on which seeds remain in an
361 area and are available to germinate and establish into the vegetative community, we encourage
362 the use of artificial grass traps because they allow natural processes such as secondary dispersal
363 and granivory to occur.

364 Here we present a methodological comparison between the seed recovery ability of
365 artificial grass traps and sticky traps for measuring seed rain in grasslands. While we find both
366 methods effectively recover seeds, ~~overall this comparison underscored that artificial grass traps~~
367 ~~are much simpler and less hassle to use~~the trap types each have their own pros and cons. Sticky
368 traps are a better measure of primary dispersal if they are replaced frequently. Artificial grass
369 traps are effective at seed recovery under a wider range of weather conditions (e.g. can capture
370 seeds post-frost), and allow for future, downstream use of seeds once the traps have been
371 removed from the field. This method will allow for coordinated understanding of seed rain across
372 grasslands (Myers and Harms, 2009; Arruda et al., 2018). Future seed rain studies will do well to
373 consider the time they leave the traps out in the field, as seeds tend to be increasingly lost
374 through time (Fig. S1). This seed loss is inevitable, and represents the fates of seeds on the soil
375 surface (Chambers and Macmahon, 1994; Forget et al., 2005), as small mammals and insects
376 consume many seeds in grasslands throughout the season (Howe et al., 2002; Peters, 2007; Kuli-
377 Révész et al., 2021). Deciding how often to replace seed rain traps can often be dictated by
378 person-power and ability to get to field sites. Therefore, we suggest replacing traps at a rate that
379 is feasible and interpreting results with caution given seeds are lost through time on all trap
380 types.

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388 Author Contributions

389 LLS and LGS designed and implemented experiment, conducted statistical analysis, and wrote
390 the manuscript, all authors helped with data collection and manuscript editing.

391 Data Availability

392 Code and data for analysis and figure generation can be found at Github
393 (~~https://github.com/Sullivan-Lab/project_CarpetMethods~~https://github.
394 ~~com/Sullivan-Lab/projects_\$CarpetMethods~~), and will be made available at Dryad upon
395 publication.

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Site	Family	Species	Seed Mass (mg)
Colorado	Agavaceae	<i>Yucca glauca</i> (Nuttall)	19.902 (±
	Fabaceae	<i>fruticosa</i>	0.518)7.543 (±
			0.243)
	Poaceae	<i>Buchloe dactyloides</i> (Nutt.)	11.785 (±
	Poaceae	Engelm. <i>Bouteloua gracilis</i>	0.555)0.540 (±
			0.073)
	Asteraceae	<i>Helianthus annuus</i> (Linnaeus)	6.750 (±
	Poaceae	<i>dactyloides</i>	0.330)11.785 (±
			0.555)
	Fabaceae	<i>Dalea purpurea</i> (Ventenat)	1.545 (± 0.060)
Missouri	Iridaceae	<i>Sisyrinchium angustifolium</i>	0.840 (± 0.044)6.750
	Asteraceae	(Miller) <i>Helianthus annuus</i>	(± 0.330)
	Asteraceae	<i>Solidago rigida</i> (Linnaeus)	0.759 (± 0.025)
	Poaceae	<i>Bouteloua gracilis</i> (Kunth) Lag. Ex	0.540 (± 0.073)0.840
	Iridaceae	Griffiths <i>Sisyrinchium angustifolium</i>	(± 0.044)
	Asteraceae	<i>Echinacea pallida</i> (Nuttall)	4.600 (± 0.183)0.215
	Asteraceae	Nuttall <i>Coreopsis tinctoria</i>	(± 0.014)
	Fabaceae	<i>Desmodium canadense</i> (Linnaeus) de Candolle in A.P. de Candolle and A.L.P.P. de Candolle	4.420 (± 0.302)
	Fabaceae	<i>Lespedeza capitata</i>	3.280 (± 0.099)4.600
	Asteraceae	(Michaux) <i>Echinacea pallida</i>	(± 0.183)
	Poaceae	<i>Sporobolus heterolepis</i> (A. Gray) A.	2.000 (± 0.087)3.280
	Fabaceae	Gray <i>Lespedeza capitata</i>	(± 0.099)
	Poaceae	<i>Schizachyrium scoparium</i> (Michx.)	1.830 (± 0.114)0.250
	Asteraceae	Nash <i>Rudbeckia hirta</i>	(± 0.010)
	Asteraceae	<i>Rudbeckia hirta</i> (Linnaeus)	0.250 (± 0.010)1.830
	Poaceae	<i>Schizachyrium scoparium</i>	(± 0.114)
	Asteraceae	<i>Coreopsis tinctoria</i> (Nuttall)	0.215 (± 0.014)
	Poaceae	<i>Sporobolus heterolepis</i>	2.000 (± 0.087)
	Campanulaceae	<i>Triodanis perfoliata</i>	0.025 (± 0.002)

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Table 1: — A list of all species used in the experiments in Colorado (CO) and Missouri

\$71 (MO). Seed mass is measured in milligrams and reported as mean and standard deviation.
\$72

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	Estimate	St. Error	z value	p value
Intercept(artificial grass)	0.589	0.185	3.18	0.001
trap type(sticky)	1.558	0.044	35.24	<0.0001
seed mass	102.518	5.683	18.04	<0.0001
time	-0.857	0.024	-36.24	<0.0001
trap type(sticky):seed mass	-115.204	7.667	-15.03	<0.0001

\$74 Table 2: — Summary table of our statistical model results from our generalized linear mixed
 575 effects model exploring how trap type, seed mass and time since deployment alters the
 576 proportion of seeds recovered.

	Estimate	St. Error	z value	p value
Intercept(post-sticky)	0.152	0.286	0.53	0.596
artificial grass	0.166	0.33	0.507	0.612
sticky	1.647	0.394	4.18	<0.0001

\$77 Table 3: — Summary table from our statistical model results from our ‘post-sticky’ analysis
 578 using a generalized linear mixed effects model exploring how trap type (post-sticky, vs artificial
 579 grass vs sticky) alters the proportion of seeds recovered.



Figure 1: Depiction of our experimental sites, seeds, and methods. A) The Colorado grassland site. B) The Missouri grassland site. C) Missouri seeds used in the experiment (from left to right: *Coreopsis tinctoria*, *Desmodium canadense*, *Echinacea pallida*, *Lespedeza capitata*, *Rudbeckia hirta*, *Schizachyrium scoparium*, *Sporobolus heterolepis*, *Triodanis perfoliata*). D) Colorado seeds used in the experiment (from left to right: *Amorpha fruticosa*, *Bouteloua gracilis*, *Buchloe dactyloides*, *Dalea purpurea*, *Helianthus annuus*, *Solidago rigida*, *Sisyrinchium angustifolium*, *Yucca glauca*). E) A comparison of the sticky and artificial grass traps after they were placed in the field. F) artificial grass traps with rabbit droppings. G) Sticky traps covered in ant bodies and leaves.

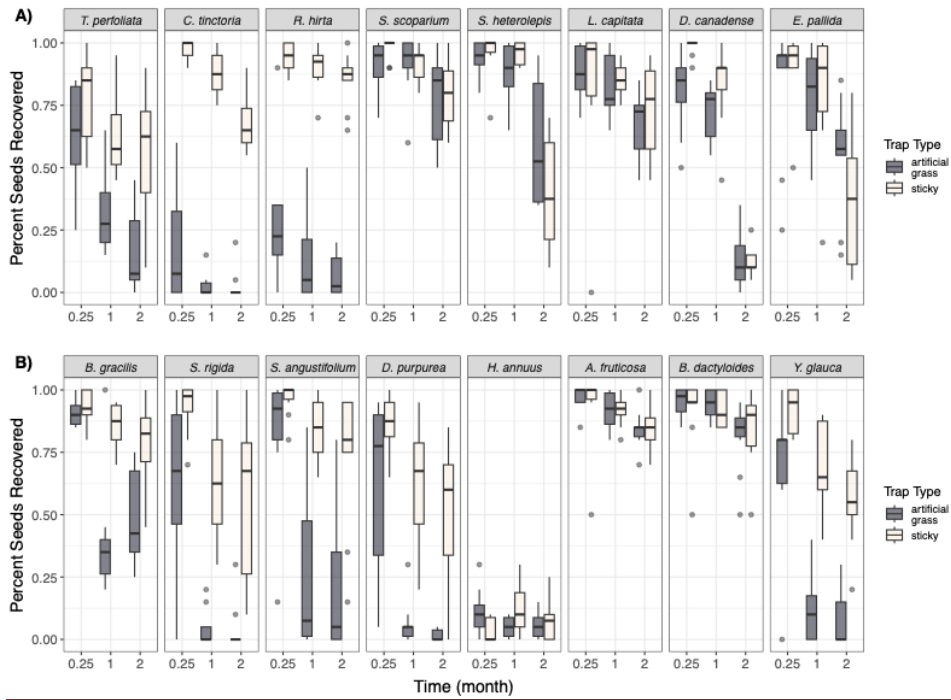
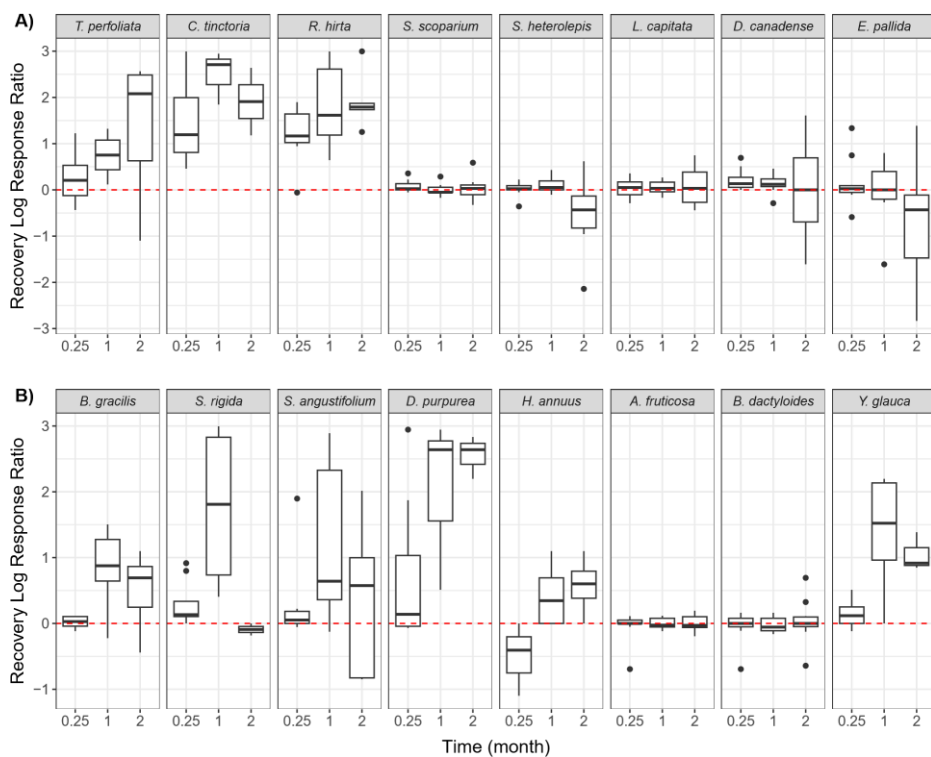
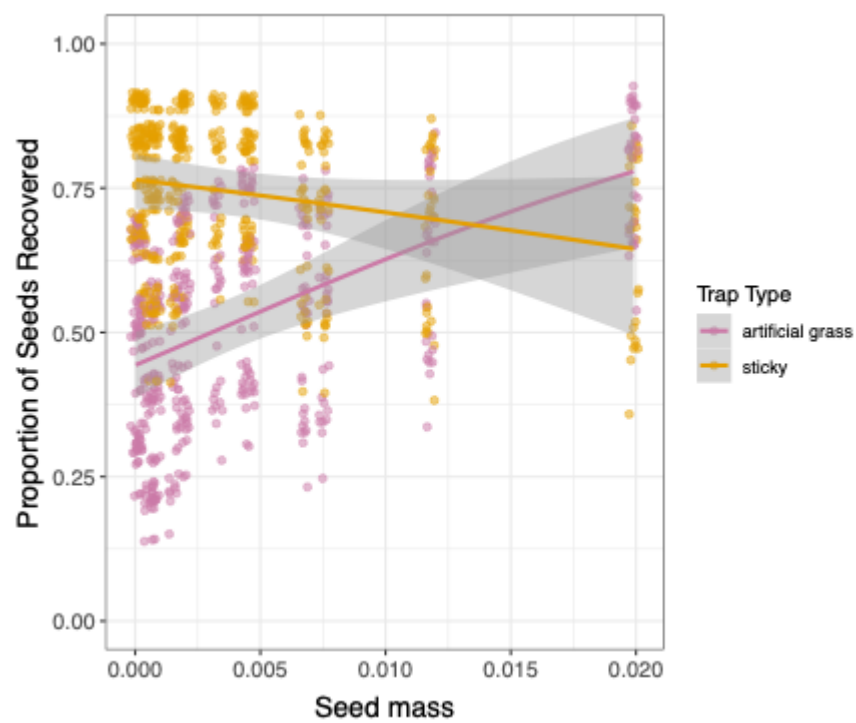


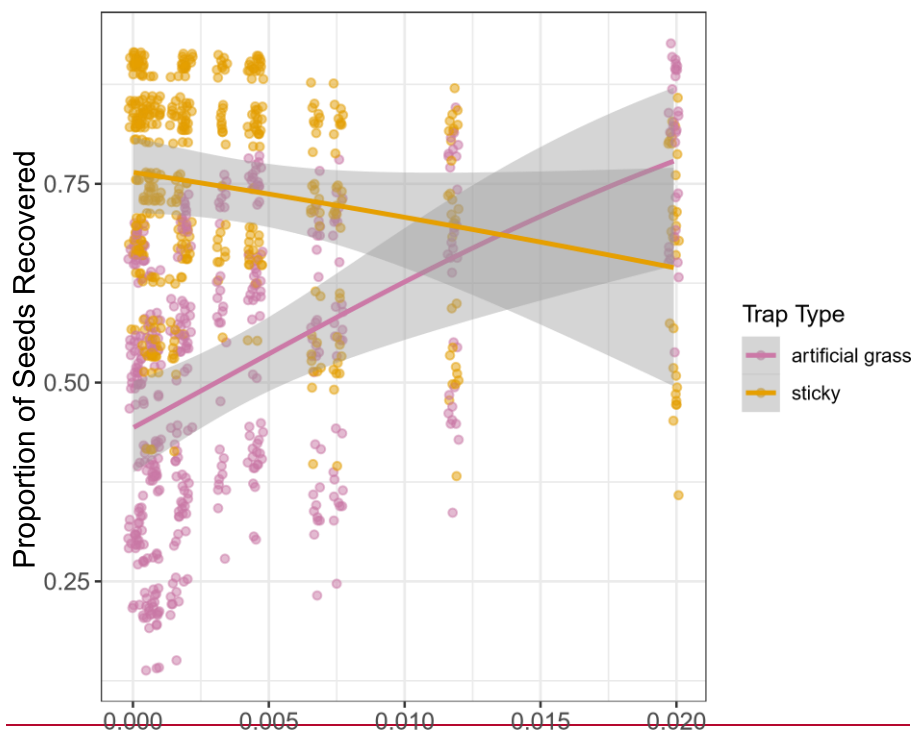
Figure 2: Proportion of seeds recovered for each species from both artificial grass (purple) and sticky (pale pink) traps at the A) Missouri site, and B) Colorado site. For each state, panels are in seed size ordered from smallest (left) to largest (right).



595

596 **Figure 3:** Log response ratio of proportion of seeds recovered in sticky traps relative to
 597 proportion of seeds recovered on artificial grass traps from the A) Missouri site, and B) Colorado
 598 site. For each state, panels are in seed size ordered from smallest (left) to largest (right). Values
 599 of zero (red dashed line) represent equal likelihood of recapture in sticky vs artificial grass traps.
 600 Values above zero indicate where seeds were more likely to be recovered on sticky traps. Values
 601 below zero represent where seeds more likely to be recovered on artificial grass traps.

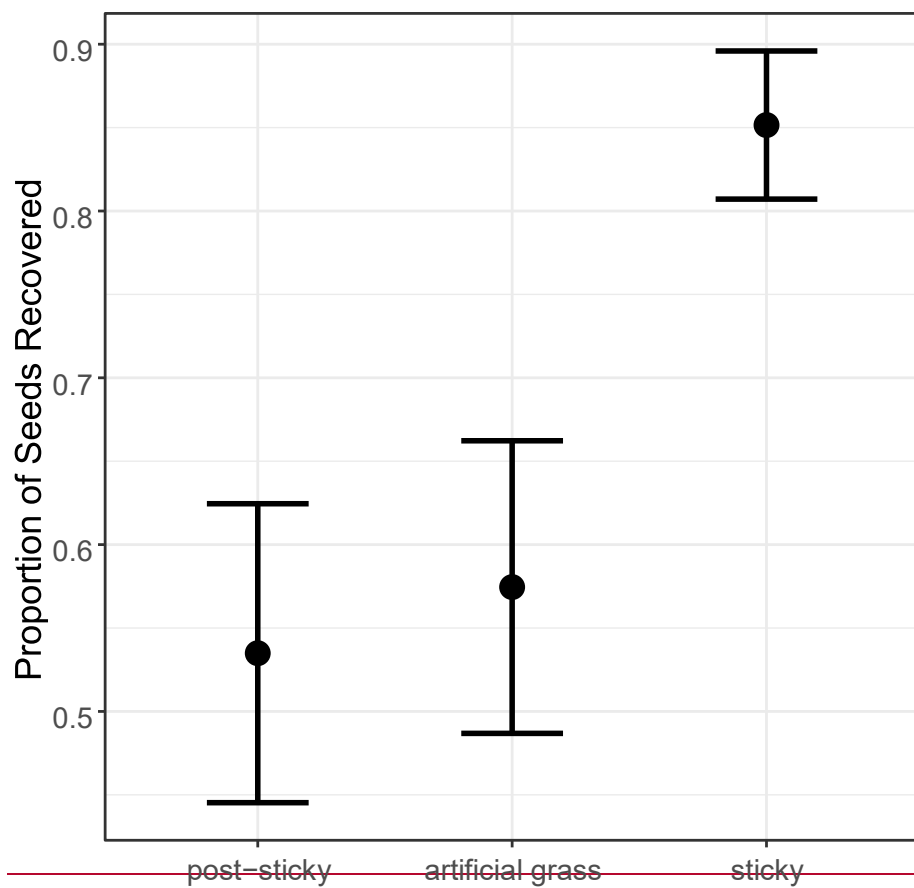




Seed mass

Figure 4: Proportion of seeds recovered in relationship to seed mass with respect to trap type (pink points are artificial grass traps, orange points are sticky traps). As seed mass increases, we find a decrease in the proportion of seeds recovered from sticky traps, and an increase in proportion of seeds recovered from artificial grass traps. Points and lines represent model predicted values, points have been jittered.

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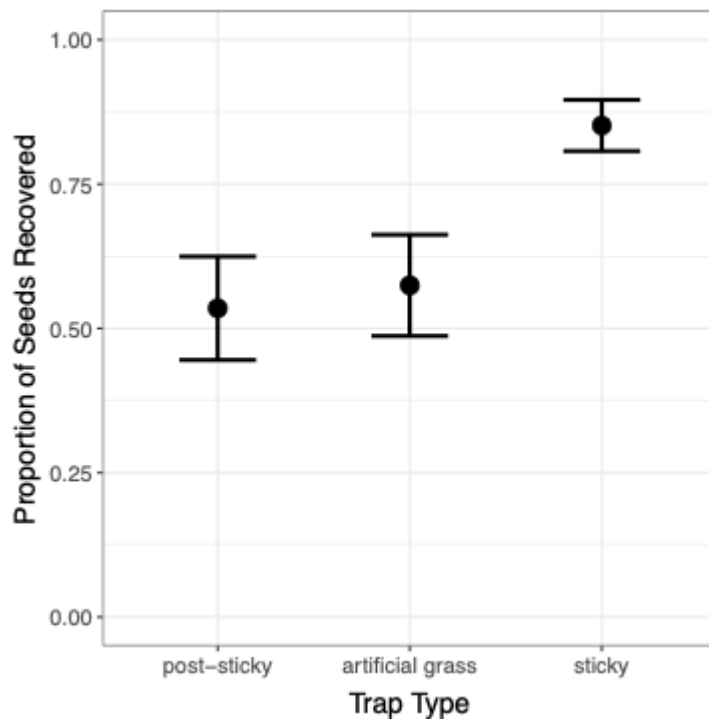


Figure 5: Proportion of seeds recovered with respect to trap type for the traps that had been out in the field for one month in Missouri (i.e. sticky, artificial grass and post-sticky traps). Data are mean model predicted values, and error bars represent standard deviation.

Running head: Sullivan et al. Grassland seed rain methods comparison

Are they two seeds in a pod? Comparing seed rain recovery using artificial grass carpets versus sticky traps in grasslands.

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Keywords: seed arrival, plant ecology, community ecology, methods, artificial grass carpet,
sticky traps, seed rain, grasslands

Type of Article: Application Article

ABSTRACT

Premise. Seed dispersal is a critical process for plant community assembly, however natural rates of seed arrival are rarely quantified when compared to other assembly mechanisms, especially in herbaceous communities.

Methods. Here we compare the utility of artificial grass carpet squares (‘artificial grass’) for capturing seed rain to classic ‘sticky trap’ methods. We placed paired sticky traps and artificial grass squares in two grassland ecosystems. We added known numbers of seeds of multiple species to each trap, and recovered seeds at one week, one month and two month intervals.

Results. We found that both trap types lost seeds through time at similar rates, but each trap type had pros and cons. Overall sticky traps retained more seeds and measured primary dispersal, but recovering seeds was difficult and hindered by debris stuck to the traps. Alternatively, artificial grass traps measured effective dispersal as seeds were lost through time to secondary dispersal and granivory, but recovered seeds could be handled easily and retained for long-term storage and germination.

Discussion. We encourage the broad adoption of seed rain studies to increase links between theoretical and empirical community ecology. Both sticky traps and artificial grass traps provide ways to measure seed rain in grasslands, but vary in the types of information they provide.

INTRODUCTION

Dispersal is a critical process that structures populations and communities, linking local patch-scale dynamics to regional landscape-level processes (Amarasekare, 2003; Snyder and Chesson, 2003; Leibold and Chase, 2018). Within plant communities, seed dispersal is one of the primary mechanisms of arrival to new habitats, allowing species to track changing environmental conditions and promoting landscape connectivity (Cubiña and Aide, 2001; Simberloff, 2009; Beckman and Sullivan, 2023). Dispersal also plays a critical role in assembly and long-term diversity patterns, such as maintaining source-sink dynamics (Mouquet and Loreau, 2003; Craig et al., 2025), allowing long-term species coexistence (Snyder and Chesson, 2003; Berkley et al., 2010), and maintaining genetic diversity in populations (Jordano, 2017). A species' dispersal ability also has direct management implications, often increasing native diversity after restoration, or alternatively, facilitating the invasion of non-native species (Von Der Lippe and Kowarik, 2007; Brudvig et al., 2009; Sperry et al., 2019). Despite its importance as a critical process for populations and communities (Schupp and Fuentes, 1995; Levine and Murrell, 2003; Vellend, 2010), dispersal is notoriously difficult to quantify empirically (Bullock et al., 2006; Beckman and Sullivan, 2023). These methodological challenges have created a gap between our theoretical understanding of the importance of dispersal versus our empirical understanding and use in applied settings (Bolker et al., 2003), making dispersal arguably the least empirically understood mechanism of community assembly (HilleRisLambers et al., 2012; Vellend, 2016).

One way to measure seed dispersal is by estimating seed rain, or the seeds that fall into a given area (Arruda et al., 2018). While this method quantifies only a single point in the dispersal trajectory of a seed (the arrival stage; Travis et al., 2012), it is arguably the most important phase to understand for plant community ecology, as it provides information on the number and type of seeds that arrive to a specific location. If seeds are recovered before post-dispersal seed predation or secondary dispersal occurs, then this is considered a measure of

primary dispersal. However, if seeds that fall to a given area are then lost, this measure of seed arrival is considered to be *effective dispersal* as it constitutes the seeds that have a chance to germinate and establish in a given area (Nathan and Muller-Landau, 2000). Quantifying seed rain as a measure of seed arrival across many different natural environments is critical for building strong links between plant community ecology experiments and theory (Myers and Harms, 2009).

Seed rain is typically measured through the use of seed traps placed on or near the ground in an ecosystem of interest (Kollmann and Goetze, 1998; Cottrell, 2004). Seed traps are commonly used in forest systems to measure seed rain (Cottrell, 2004; Rogers et al., 2017; Piotto et al., 2019; Huanca Nuñez et al., 2021), and when there are known parents, dispersal trajectories can be calculated using inverse modeling (Nathan and Muller-Landau, 2000; Wälder et al., 2009). These traps are considered to measure “seed capture”, as traps are empty when placed in the field, and seeds that fall into the traps from the surrounding community are captured and measured when traps are removed. Seed rain measurements, however, are much less common in ecosystems with predominantly herbaceous communities (e.g. grasslands, tundra), with only a handful of studies estimating seed rain in grasslands specifically (Rabinowitz and Rapp, 1980; West and Durham, 1991; Schott and Hamburg, 1997; Kettenring and Galatowitsch, 2011; Willand et al., 2015; Wynne et al., 2024). This dearth of knowledge on herbaceous seed rain likely occurs because capturing small herbaceous seeds is challenging (Myers and Harms, 2009) and seed identification resources for even common species’ can be scarce (Wynne et al., 2024). The main method for capturing seed rain in grasslands is via sticky traps, where a flat surface (often a Petri dish) is sprayed with a sticky substance, and seeds that falls onto the surface are captured and later recorded (Rabinowitz and Rapp, 1980). Sticky traps present multiple practical problems, including that they need to be replaced frequently because the sticky surface can become completely covered in leaves and debris, thereby blocking newly fallen seeds from capture. Furthermore, sticky traps are no longer sticky once temperatures fall below freezing,

despite this being a critical time for seed dispersal in temperate grasslands (Rabinowitz and Rapp, 1980). Lastly, using captured seeds from sticky traps for additional work, such as germination trials or trait measurement, is rarely possible as seeds become coated with the sticky residue.

An alternative method for measuring seed rain is to use artificial grass carpets. With this method, artificial grass patches are fixed to the ground, capturing seeds that fall and become lodged in the artificial grass strands. This method has previously been used to sample seed rain in the Arctic (Ulf Molau, 1996; Molau and Larsson, 2000), and provides a potential solution to the many issues that arise with collecting seed rain via sticky traps in grasslands more generally. Artificial grass can remain in the field for long periods of time and still capture seeds, withstand large temperature and hydrologic fluctuations, and collected seeds can be germinated after capture should visual identification not be possible due to lack of seed identification resources (Ulf Molau, 1996; Wolters et al., 2004). These features make artificial grass traps a viable option for seed rain studies in grasslands. For example, recently Wynne et al. (2024) used artificial grass carpets for capturing seed rain in a successional chronosequence of temperate tallgrass prairies and were able to identify ~27,000-124,000 seeds per square meter depending on the age of the grassland. Although the artificial grass carpet method appears promising for widespread use for collecting seed rain in grasslands, we currently lack a comparison between this and the sticky trap method to compare their ability to capture seeds and to consider potential biases across methods.

Here, we investigate the effectiveness of traditional sticky traps (Tangle-Trap® resin on Plexiglass® Petri plates) as compared to artificial grass traps for measuring seed rain in grasslands. While our ultimate goal is to use these traps to capture seed rain in the wild, we first need to quantify how well each trap type retains known numbers of seeds under field-conditions. Thus, we conduct “seed recovery” experiments where we put known numbers of seeds on each trap type and see how well we are able to recover these same seeds at

distinct time intervals. This recovery measure provides information on how confident we can feel in whether we are measuring *primary* or *effective* seed dispersal. If we recover most seeds, we consider this primary dispersal. If we recover less seeds, we consider this effective dispersal. We examined the seed retention efficacy of both methods through time, comparing the proportion of seeds recovered across two temperate grassland ecosystems, a diversity of species, and a range of seed sizes. While traditional sticky traps had a higher recovery rate of smaller seeds, both methods recovered a decently high proportion of seeds overall, and seeds from artificial grass traps were in better condition for counting, identification, and post-processing measurements. As such, we suggest that sticky trap capture data are more likely to measure primary dispersal, whereas artificial grass trap capture data are more likely to measure effective dispersal. By having a robust comparison of different seed rain capture methods, we can better advance our empirical understanding of dispersal, narrowing the current gap between theory and data, and advancing our understanding of the role of dispersal in grassland restoration and conservation.

METHODS

Study sites

We replicated experimental tests of seed capture methods for detecting seed rain in two distinct grassland communities: a shortgrass prairie in northern Colorado, and a tallgrass prairie in central Missouri (Fig. 1A, B). The Colorado (CO) study occurred on private land bordering the Cathy Fromme Prairie Open Space, a 267 ha remnant of native shortgrass prairie. The site is relatively flat, surrounded by mature cottonwood trees, and receives approximately 38 cm of precipitation annually (Balok et al., 1995). The Missouri (MO) study occurred on a restored tallgrass prairie owned by the University of Missouri. The prairie was located on campus, and is

surrounded by buildings and foot traffic, and receives approximately 108 cm of annual precipitation (Gu et al., 2015).

Experimental design

We tested the effectiveness of sticky traps versus artificial grass carpets for seed retention through time. Following standard protocol (Rabinowitz and Rapp, 1980; Kollmann and Goetze, 1998; Chabrierie and Alard, 2005), sticky traps consisted of 10 cm x 10 cm Plexiglass plates coated with Tangle-Trap® sticky coating. To compare this standard method to artificial grass traps, we used 10 cm x 10 cm synthetic lawn turf squares (TrafficMASTER® TruGrass Emerald Artificial Turf) with blade heights between 1.5 to 2.5 cm tall.

At the two sites, we replicated 10 blocks of paired sticky and artificial grass traps that we collected 1 week, 1 month, and 2 months after deployment to examine seed recovery through time. We initially deployed all traps on July 10, 2019 in Colorado and July 17, 2019 in Missouri. Each block consisted of three sticky traps and three artificial grass traps for a total of 10 replicates of each trap type for each time point and each site. Within each block, paired traps were placed side by side, closer to each other than the replicates in other blocks. When deploying traps, we removed any litter or vegetation directly under the traps so that they were placed directly on the soil surrounded by natural vegetation (Fig. 1E). We used two ground staples to secure opposite sides of each trap. On each trap, we then placed 20 seeds of 8 locally occurring species that were not found in the neighboring vegetative community at our study sites. Seed mixes were chosen such that they encompassed a range of functional groups and seed mass (see Table 1 for species and associated seed mass, Fig. 1C, D). At the Colorado site, we used only 10 seeds for the species at the highest end our seed mass range, *Yucca glauca*, due to their large size (~7x10mm, Fig. 1D), resulting in a total of 150 (CO) and 160 (MO) seeds per trap.

We collected paired sticky and artificial grass traps 1 week, 1 month, and 2 months after deployment. All traps were taken back to the lab for processing. We recorded the total number of

seeds of each focal species that were recaptured on each trap, and noted if seed fragments were evident to suggest seed herbivory or animal activity.

At the Missouri site, we also wanted to test the efficacy of the sticky traps after they had been in the field for some time and accumulated leaves and other debris. Therefore, at the initial trap deployment date, we placed an additional sticky trap at each block that was coated in Tangle-Trap®, but did not have the seed mix added initially. After one month of sitting in the grassland and accumulating leaves and other debris naturally, we added the same seed mix as above to these traps, and then collected these traps along with the two month traps. We termed this treatment, the ‘post-sticky’ treatment. We recovered and identified seeds from these traps in the same manner as mentioned above. This allowed us to more closely examine how well sticky traps recovered seeds after longer periods of time in the field.

Statistical analysis

We explored both absolute and relative seed recovery. To visualize how relative seed recovery varied across species and trap type, we calculated log response ratios of seed recovery of sticky traps relative to artificial grass traps for each species at each site (i.e. the log of % recovery on sticky traps divided by % recovery on artificial grass traps). For this formulation, values greater than zero indicate seeds are more likely to be recovered on sticky traps, values of less than zero are more likely to be recovered on grass traps, and values of zero indicate seeds are equally likely to be recovered on both traps. This allows for ease of visualization of which trap type tends to recover relatively more seeds of a given species at a given time.

To determine how absolute seed recovery depended on the trap type, seed mass and the amount of time the traps remained in the field, we used generalized linear mixed effects models using the ‘glmmTMB’ package (Brooks et al., 2017) in R v4.3.3. (R Core Team, 2024). The percent of recovered seeds was our response variable, and accordingly we used a binomial model

with weights specified as the number of seeds placed on the traps to account for differing numbers of seeds per species used in the experiment. Our predictor variables included the interaction of trap type, seed mass and time in the field. We also included an additive random intercept for block nested within site (MO or CO). We used a backward selection approach to determine which interactions we kept in the model, and removed non-significant interactions when they arose. We calculated contrasts between trap type predictions for given seed masses using ‘emmeans’ (Lenth, 2023).

To determine how well the sticky traps recovered seeds once the traps had been in the field for some time and acquired debris, we explored how recovery differed between the poststicky treatment and the other trap types that contained seeds for one month. At the Missouri site only, we subset our data to all traps that had seeds on them for one month (artificial grass, sticky and post-sticky). We then ran a similar generalized linear mixed effects model as above, with our response variable as the proportion of seeds recovered (modeled as a binomial distribution), our predictor variable as trap type, and two additive random intercepts, one for block and one for species. We then calculated contrasts with ‘emmeans’ between trap type.

We wrangled and visualized data with the ‘tidyverse’ package (Wickham et al., 2019), and ran all code in R Studio v2024.12.0+467 (Posit Team, 2024).

RESULTS

Absolute and relative seed recovery varied by species and time (Figs. 2,3). There was high variability among absolute recovery by species, with some species having higher recovery broadly (e.g. *S. scoparium*, *B. dactyloides*), others had low recovery across all trap types (e.g. *H. annuus*), but most falling somewhere in the middle. We also observed that some species were more likely to be recovered on sticky traps (positive log response ratio, e.g. *C. tinctoria*, *D.*

purpurea), others were more likely to be recovered on artificial grass traps (negative log response ratio, e.g. *H. annuus*, *S. heterolepis*), and many were equally likely to be recovered on either trap type (e.g. *S. scoparium*, *B. dactyloides*). These trends often varied with time.

We next explored how trap type, seed mass, and the time the traps spent in the field altered the absolute proportion of seeds recovered. We found a significant interaction between seed mass and trap type, and a significant additive effect of time (Table 2). On average, sticky traps recovered more seeds than artificial grass traps, with sticky traps recovering $\sim 74\% \pm 13\%$ SD of seeds, and artificial grass traps recovering $\sim 52\% \pm 18\%$ SD of seeds. However, this relationship depended on seed mass (Fig. 4). As seed mass increased, the proportion of seeds recovered increased on artificial grass traps, but decreased on sticky traps. For small seed masses, artificial grass traps caught significantly fewer seeds (log odds estimate at seed mass of 0, estimate = -1.56 ± 0.04 SE, $p < 0.001$). However, at large seed sizes, artificial grass traps caught significantly more seeds (log odds estimate at seed mass of 0.02, estimate = 0.75 ± 0.13 SE, $p < 0.001$). Additionally, the longer the traps were in the field, the more seeds were lost (Fig. S1). Across both trap types, $\sim 30\%$ seeds were lost for each month the traps spent in the field. This loss was due in-part to seed predation as we noted seed fragments left on both trap types post-consumption.

Finally, to determine how well the sticky traps continue to capture seeds after some time in the field, we conducted the ‘post-sticky’ trial at the Missouri site. We found that sticky traps caught a significantly larger proportion of seeds than the post-sticky treatment, but there was no difference found between the seeds captured with the artificial grass and the post-sticky treatments (Fig. 5, Table 3). The post-sticky traps captured $\sim 30\%$ less seeds than the sticky traps.

DISCUSSION

A broader understanding of the natural rates of seed rain in grasslands is necessary to understand the role of seed arrival for the assembly and reassembly of plant communities under global change (Myers and Harms, 2009; Arruda et al., 2018). Standardized methods for estimating seed arrival and dispersal dynamics that are flexible and can adequately recover seeds under a myriad of conditions is critical, given that grasslands span $\sim 40\%$ of the Earth's surface, and occur on all continents across a wide range of precipitation and temperature regimes (Buisson et al., 2022). Here we provide a test of the seed recovery efficacy of two methods for measuring seed rain in grassland communities: artificial grass traps and sticky traps. These methods are both easy to deploy because of their size and minimal impact on the local community (as compared to other funnel or pitfall-based methods; Kollmann and Goetze, 1998). While both methods are successful for recovering seeds, our experimental comparison highlights that sticky traps are more likely to represent primary dispersal because they lose fewer seeds through time, whereas the artificial grass traps represent effective dispersal because fewer seeds remained at each point and thus had fewer opportunities for future germination. Obtaining estimates of local primary and effective dispersal across a range of grassland communities will help bridge the gap between dispersal theory and empiricism (Amarasekare, 2003; Bolker et al., 2003; Myers and Harms, 2009).

We find that both the artificial grass and sticky trap methods have high rates of seed recovery (72% for sticky traps on average, compared to 52% for artificial grass traps), with decreasing recovery through time ($\sim 30\%$ seeds lost per month overall). Seed loss from both trap types was due to both abiotic and biotic reasons, such as strong rainfall events washing seeds off the artificial grass or herbivory of seeds. In our study, we found evidence for both scenarios; for example a large, single day rainfall event decreased seed recovery at the Colorado site between time point 1 (1 week after deployment) and time point 2 (1 month after deployment). Evidence

of both insect and small mammal herbivory occurred across both sites, with hundreds of ants on some sticky traps (Fig. 1G), cottontail rabbit (*Sylvilagus sp.*) feces on artificial grass (Fig. 1F), and husks of seeds post-consumption found on both trap types, especially after longer periods of time. These findings indicate that no trap type is a perfect representation of *primary dispersal*, but the sticky traps come the closest. Alternatively, the artificial grass traps represents a measure of *effective dispersal* (Nathan and Muller-Landau, 2000), where recovered seeds are those that have escaped granivory or secondary dispersal and could potentially germinate and contribute to the aboveground community. Therefore, the artificial grass trap method arguably represents more “ecologically relevant dispersal” for the local community, because the traps allow for seed loss that would occur naturally on the soil surface - leading to recovered seeds being those that have the potential to germinate. We suggest future studies examine herbivory rates on artificial grass traps, deploying paired traps in herbivore removal and control plots to estimate the levels of biotic seed removal. Artificial grass trap methods could be extended from our current focus on seed rain to examine herbivory rates or even conduct preference experiments in grasslands (Edwards and Crawley, 1999; Bakker and Olff, 2003).

While sticky traps are more commonly used in field studies (e.g. Rabinowitz and Rapp, 1980) and retain more seeds on average, our results demonstrate these traps have some drawbacks. Sticky traps capture everything that comes in contact with the surface of the plate (Fig. 1G), and thus sorting seeds from plant and soil debris as well as insect bodies can be difficult. Recovery is especially difficult for small seeds that have the potential to stick to debris in the trap and be easily missed. In our work we were focused on finding a known set of species. However, once the identity of seeds are unknown (as in classic seed rain capture studies, compared to our comparative methods-focused recovery approach), the ability to lose seeds due to accidentally sticking to other material will decrease seed capture. We also noticed that some species with thinner seed coats changed color after being in contact with Tangle-Trap® for long periods of time, which can hinder identification if seeds are not removed and identified

quickly. Given how time-consuming seed capture and identification can be (Wynne et al., 2024), it is critical to have a method that allows samples to be stored for long periods and remain unaffected. In addition, it is nearly impossible to remove seeds from sticky traps and then germinate them to determine viability or identity, whereas this is quite simple with artificial grass traps. The potential to germinate ‘unknowns’ is, in our view, a key advantage of artificial grass traps, given unknowns are extremely likely in seed rain studies (Wynne et al., 2024). Finally, the timing of sticky traps is extremely important for their effectiveness. Sticky traps no longer capture seeds effectively once they have been frozen (Ulf Molau, 1996), which makes post-frost seed rain collection impossible despite many temperate seeds dispersing late in the growing season or under snowy conditions (Rabinowitz and Rapp, 1980; Molau and Larsson, 2000). In addition, our results from the Missouri site clearly show that sticky traps work best when they are in the field for short periods of time, or else debris collects on the trap and hinders seed capture and recovery (Fig. 5). This is in comparison to artificial grass traps that continue to effectively collect and store seeds regardless of time or temperature.

Our results demonstrated that the relationship between seed mass and seed recovery changes with trap type (Fig. 4). As seed size increases, we see a small decrease in the proportion of seeds recovered on sticky traps, but a large increase in the proportion of seeds recovered on artificial grass traps. This result likely arises from differences in the body size of the granivores consuming seeds from the traps. Broadly, there is a positive relationship between granivore body size and the size of seeds they consume (Muñoz and Bonal, 2008), with smaller-bodied insect granivores tending to favor smaller seeds than larger-bodied small mammal granivores (Rey et al., 2002; Radtke, 2011; Vaz Ferreira et al., 2011; Maron et al., 2012). The loss of smaller seeds from the artificial grass traps likely reflects how well the Tangle-Trap® captures insects like ants that attempt to remove seeds from these traps, but instead become stuck to the surface of the trap (Fig. 1G). Therefore, more smaller seeds remain to be recaptured on the sticky traps because insect granivores are not removing them. Conversely, larger seeds are more

likely to be recovered on the artificial grass traps. Here, larger seeds are likely consumed by small mammals, and are likely easier to consume from sticky traps because they are fixed on the soil surface and not hidden among the strands of grass as they are in the artificial grass traps. This decrease in handling time could increase consumption rates of larger seeds from sticky traps. While the exact reason for why we find this interaction between trap type and seed mass should be further explored, this result underscores the importance of considering the difference in the results provided by the different trap methods. If the goal is to understand primary seed dispersal and to quantify the number and type of seeds that fall regardless of their fate, we encourage the use of sticky traps that are replaced very frequently (and of course considering all possible concerns listed above). However, if the goal is to understand effective dispersal, which focuses more on which seeds remain in an area and are available to germinate and establish into the vegetative community, we encourage the use of artificial grass traps because they allow natural processes such as secondary dispersal and granivory to occur.

Here we present a methodological comparison between the seed recovery ability of artificial grass traps and sticky traps for measuring seed rain in grasslands. While we find both methods effectively recover seeds, the trap types each have their own pros and cons. Sticky traps are a better measure of primary dispersal if they are replaced frequently. Artificial grass traps are effective at seed recovery under a wider range of weather conditions (e.g. can capture seeds post-frost), and allow for future, downstream use of seeds once the traps have been removed from the field. This method will allow for coordinated understanding of seed rain across grasslands (Myers and Harms, 2009; Arruda et al., 2018). Future seed rain studies will do well to consider the time they leave the traps out in the field, as seeds tend to be increasingly lost through time (Fig. S1). This seed loss is inevitable, and represents the fates of seeds on the soil surface (Chambers and Macmahon, 1994; Forget et al., 2005), as small mammals and insects consume many seeds in grasslands throughout the season (Howe et al., 2002; Peters, 2007; Kuli-Révész et al., 2021). Deciding how often to replace seed rain traps can often be dictated by person-power

and ability to get to field sites. Therefore, we suggest replacing traps at a rate that is feasible and interpreting results with caution given seeds are lost through time on all trap types.

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Author Contributions

LLS and LGS designed and implemented experiment, conducted statistical analysis, and wrote the manuscript, all authors helped with data collection and manuscript editing.

Data Availability

Code and data for analysis and figure generation can be found at Github (https://github.com/Sullivan-Lab/project_CarpetMethods), and will be made available at Dryad upon publication.

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Site	Family	Species	Seed Mass (mg)
Colorado	Agavaceae	<i>Yucca glauca</i> (Nuttall)	19.902 ($\pm$ 0.518)
	Poaceae	<i>Buchloe dactyloides</i> (Nutt.) Engelm.	11.785 ($\pm$ 0.555)
	Asteraceae	<i>Helianthus annuus</i> (Linnaeus)	6.750 ($\pm$ 0.330)
	Fabaceae	<i>Dalea purpurea</i> (Ventenat)	1.545 ($\pm$ 0.060)
	Iridaceae	<i>Sisyrinchium angustifolium</i> (Miller)	0.840 ($\pm$ 0.044)
	Asteraceae	<i>Solidago rigida</i> (Linnaeus)	0.759 ($\pm$ 0.025)
	Poaceae	<i>Bouteloua gracilis</i> (Kunth) Lag. Ex Griffiths	0.540 ($\pm$ 0.073)
Missouri	Asteraceae	<i>Echinacea pallida</i> (Nuttall) Nuttall	4.600 ($\pm$ 0.183)
	Fabaceae	<i>Desmodium canadense</i> (Linnaeus) de Candolle in A.P. de Candolle and A.L.P.P. de Candolle	4.420 ($\pm$ 0.302)
	Fabaceae	<i>Lespedeza capitata</i> (Michaux)	3.280 ($\pm$ 0.099)
	Poaceae	<i>Sporobolus heterolepis</i> (A. Gray) A. Gray	2.000 ($\pm$ 0.087)
	Poaceae	<i>Schizachyrium scoparium</i> (Michx.) Nash	1.830 ($\pm$ 0.114)
	Asteraceae	<i>Rudbeckia hirta</i> (Linnaeus)	0.250 ($\pm$ 0.010)
	Asteraceae	<i>Coreopsis tinctoria</i> (Nuttall)	0.215 ($\pm$ 0.014)

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Table 1: A list of all species used in the experiments in Colorado (CO) and Missouri (MO). Seed mass is measured in milligrams and reported as mean and standard deviation.

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	Estimate	St. Error	z value	p value
Intercept(artificial grass)	0.589	0.185	3.18	0.001
trap type(sticky)	1.558	0.044	35.24	<0.0001
seed mass	102.518	5.683	18.04	<0.0001
time	-0.857	0.024	-36.24	<0.0001
trap type(sticky):seed mass	-115.204	7.667	-15.03	<0.0001

535 Table 2: Summary table of our statistical model results from our generalized linear mixed effects
 536 model exploring how trap type, seed mass and time since deployment alters the proportion of
 537 seeds recovered.

	Estimate	St. Error	z value	p value
Intercept(post-sticky)	0.152	0.286	0.53	0.596
artificial grass	0.166	0.33	0.507	0.612
sticky	1.647	0.394	4.18	<0.0001

538 Table 3: Summary table from our statistical model results from our ‘post-sticky’ analysis using a
 539 generalized linear mixed effects model exploring how trap type (post-sticky, vs artificial grass vs
 540 sticky) alters the proportion of seeds recovered.



Figure 1: Depiction of our experimental sites, seeds, and methods. A) The Colorado grassland site. B) The Missouri grassland site. C) Missouri seeds used in the experiment (from left to right: *Coreopsis tinctoria*, *Desmodium canadense*, *Echinacea pallida*, *Lespedeza capitata*, *Rudbeckia hirta*, *Schizachyrium scoparium*, *Sporobolus heterolepis*, *Triodanis perfoliata*). D) Colorado seeds used in the experiment (from left to right: *Amorpha fruticosa*, *Bouteloua gracilis*, *Buchloe dactyloides*, *Dalea purpurea*, *Helianthus annuus*, *Solidago rigida*, *Sisyrinchium angustifolium*, *Yucca glauca*). E) A comparison of the sticky and artificial grass traps after they were placed in the field. F) artificial grass traps with rabbit droppings. G) Sticky traps covered in ant bodies and leaves.

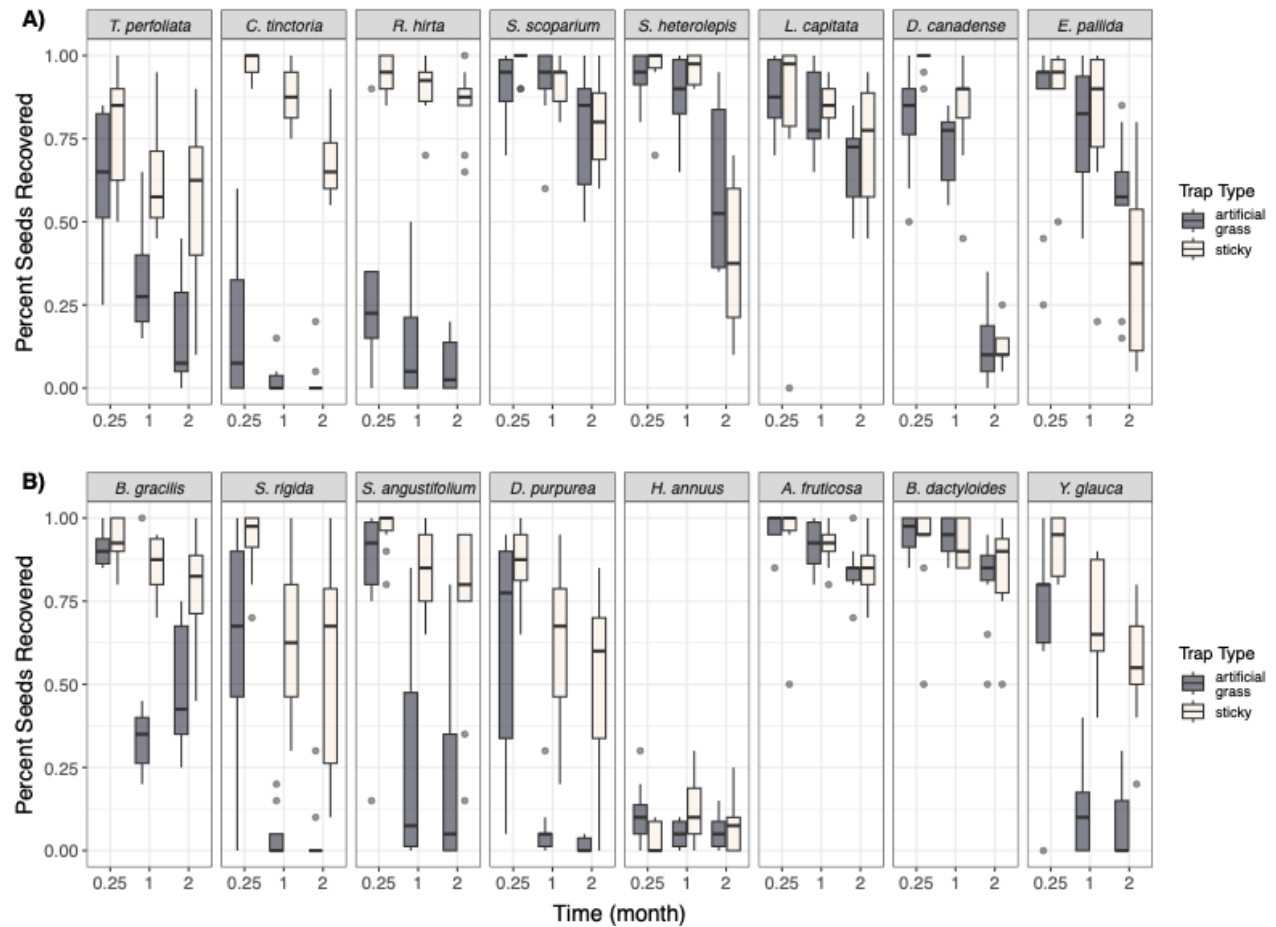


Figure 2: Proportion of seeds recovered for each species from both artificial grass (purple) and sticky (pale pink) traps at the A) Missouri site, and B) Colorado site. For each state, panels are in seed size ordered from smallest (left) to largest (right).

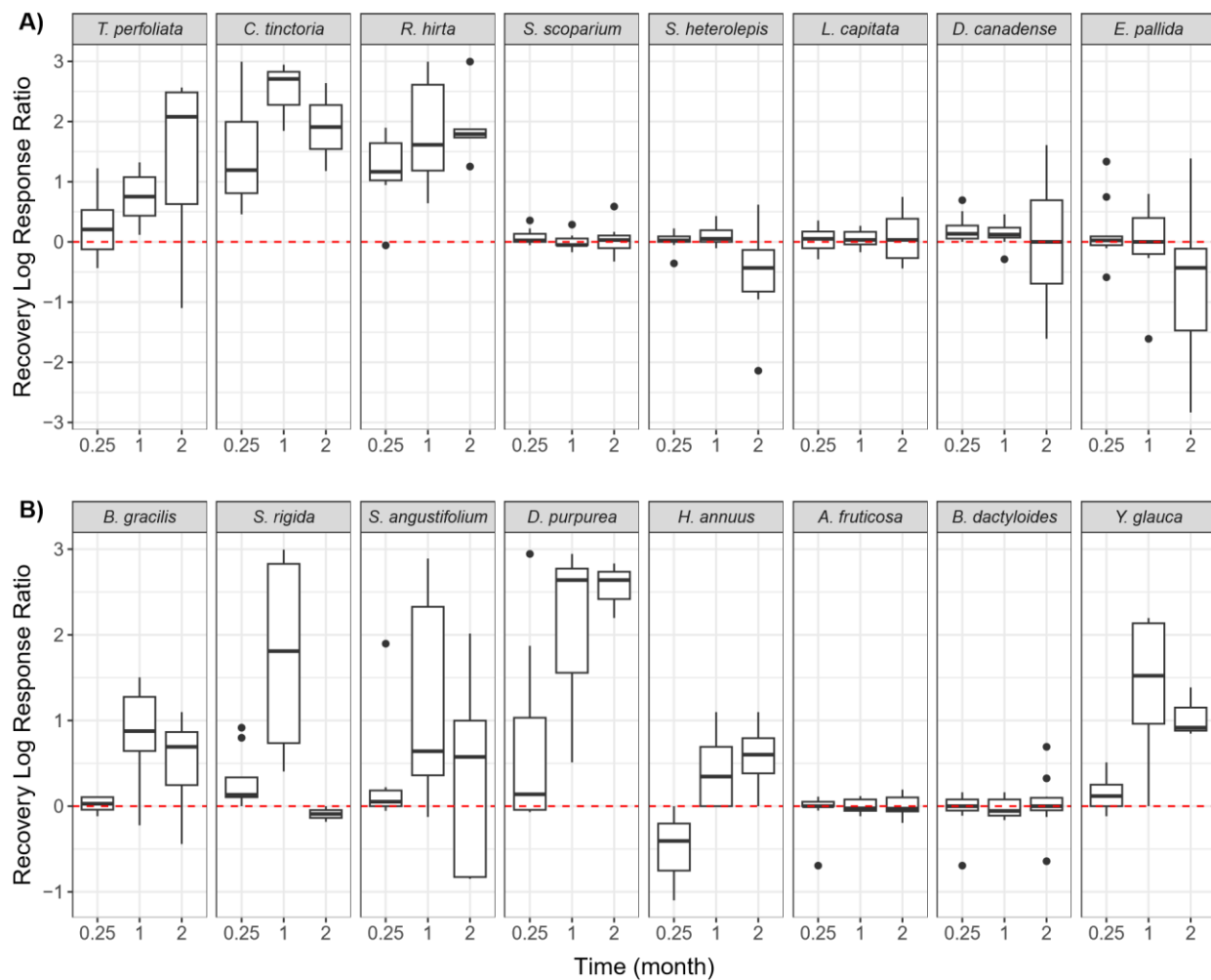


Figure 3: Log response ratio of proportion of seeds recovered in sticky traps relative to proportion of seeds recovered on artificial grass traps from the A) Missouri site, and B) Colorado site. For each state, panels are in seed size ordered from smallest (left) to largest (right). Values of zero (red dashed line) represent equal likelihood of recapture in sticky vs artificial grass traps. Values above zero indicate where seeds were more likely to be recovered on sticky traps. Values below zero represent where seeds more likely to be recovered on artificial grass traps.

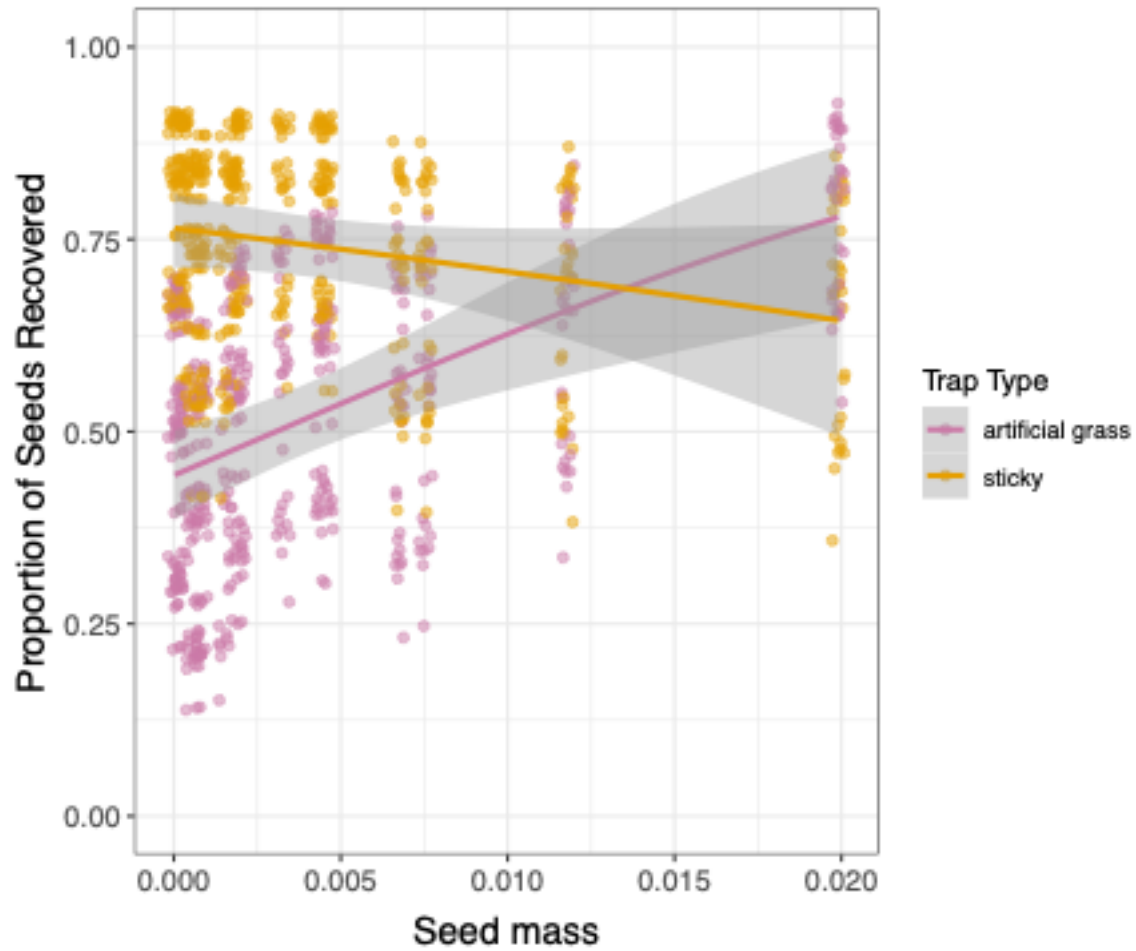
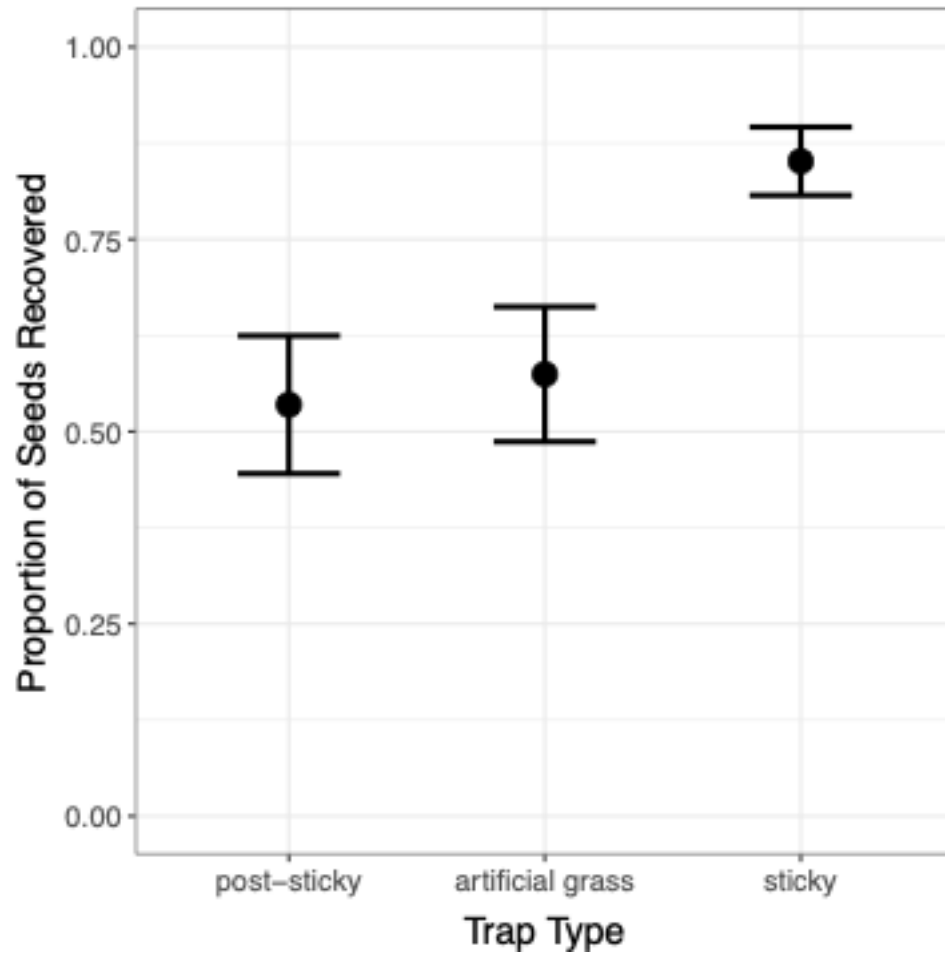
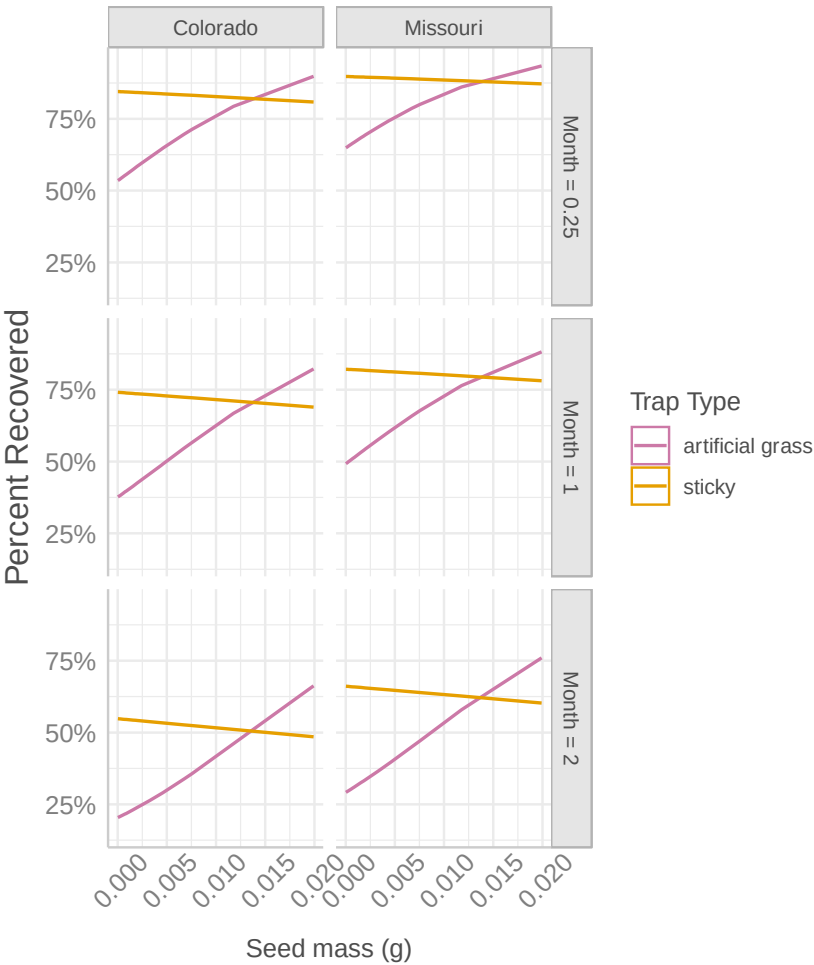


Figure 4: Proportion of seeds recovered in relationship to seed mass with respect to trap type (pink points are artificial grass traps, orange points are sticky traps). As seed mass increases, we find a decrease in the proportion of seeds recovered from sticky traps, and an increase in proportion of seeds recovered from artificial grass traps. Points and lines represent model predicted values, points have been jittered.



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571 Figure 5: Proportion of seeds recovered with respect to trap type for the traps that had been out in
572 the field for one month in Missouri (i.e. sticky, artificial grass and post-sticky traps). Data are
573 mean model predicted values, and error bars represent standard deviation.



Supplemental Appendix Figure 1: Predicted values for all factors in the model – the relationship of percent of seeds recovered relative to seed mass across both sites and all three times.



Potential cover image caption: Comparison of two methods for measuring seed rain in grassland ecosystems (traditional sticky trap method on the left, and artificial grass method on the right).